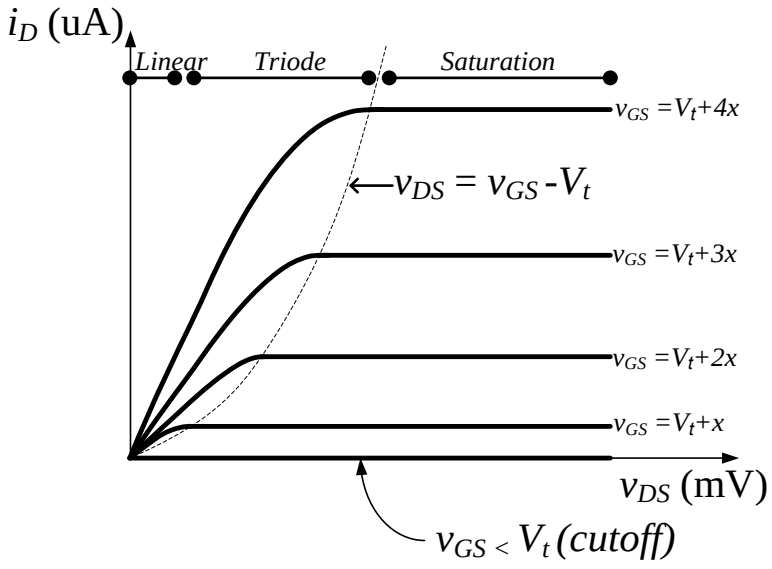
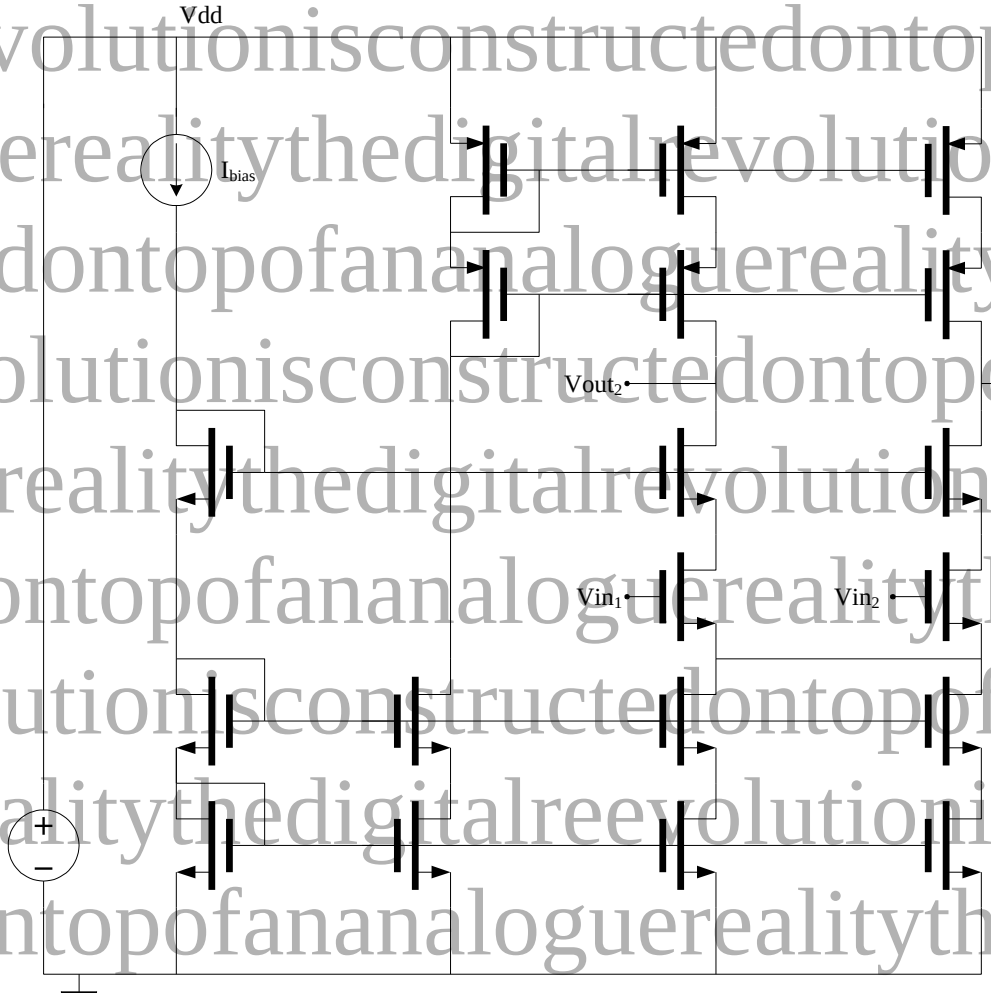
School of Electrical, Electronic and Mechanical Engineering

Analogue Integrated Circuits

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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Analogue Integrated Circuit Design

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[Chapter 4 is the most relevant to CMOS design]

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1. Introduction and Background Material

1.1.History, Context and Rational

1.1.1. The Invention of the Integrated Circuit

In July 1958 Jack Kilby of Texas Instruments conceived that a single block of germanium or silicon could be a platform for transistors, diodes, resistors and (junction) capacitors concurrently.

To insulate the various components from each other Kilby etched the silicon, in some areas all the way through. To connect them together he used gold wires resulting in a cumbersome but functional circuit. The resulting patent included the claim that interconnection could also be made by laying down a layer of gold.

Almost at the same time, a similar but far more elegant idea occurred to Robert Noyce at Fairchild in California. His work centred on the removal of the necessity to cut the die into separate components and then reconnect them at board level. Noyce visualised capacitors and resistors made in silicon. The idea of using wires had no chance in Noyce's mind, it would have simply been too expensive. But he saw that, in the planar process, this problem was already solved: the aluminium layer used to for the bonding of components to the package wires could also be used between the components.

In 1959 Noyce entered his idea into his notebook and filed for a patent application. Kilby's and Noyce's patent applications were clearly in interference and a bitter battle between the two companies started in the courts. Texas Instruments won because Kilby’s application mentioned a thin film of gold, thus seemingly anticipating Noyce. Fairchild appealed.

While the two patents were fought over in the courts, neither TI nor Fairchild could collect any royalties for integrated circuits, which were already showing explosive growth. So the two companies came to an agreement, declaring Kilby and Noyce co-inventors of the integrated circuit.

Shortly after this the appeals court handed down its decision: Noyce, not Kilby, was declared the inventor of the IC.

Setting this aside, the most important contributor to the invention of the IC was clearly Jean Hoerni with his planar process, for which he has never been adequately recognized. The planar process rates as one of the great inventions of the 20th century.

Robert Noyce died in 1990 at age 62. In 2000 Jack Kilby won the Nobel Prize for the invention of the integrated circuit

1.1.2. Why Analogue? Why CMOS?

Many electronic products are based on the application of digital technology; phones, television, video disks, hearing aids, motor controls, audio amplifiers, toys, printers etc. so one may not think that the analogue circuit features in this digital world.

Consider a DVD player: it has more analogue content than the (analogue) VCR ever did. The question is, ‘why?’.

Setting philosophy aside, the answer is about us and our interaction with our surroundings: the world is fundamentally analogue. Hearing, vision, taste, touch, smell are all analogue, so is lifting and walking. Generators, motors, loud-speakers,

microphones, solenoids, batteries, antennas, lamps, LEDs, laser diodes, sensors are fundamentally analogue components.

The digital revolution is constructed on top of an analogue reality.

This fact simply won't go away. Somewhere, somehow you have to get into and out of the digital system and connect to the real world.

In integrated circuits, the relentless pressure for faster digital speed has resulted in ever-decreasing supply voltages, which hinder high-performance analogue design. At 350nm (3.3V) there is still enough headroom for a high-performance analogue design, though 5 Volts would be better. At 180nm (1.8V) the job becomes elaborate and time-consuming and performance starts to suffer. At 120nm (1.2V) analogue design becomes very difficult even with reduced performance. At 90nm, analogue design is all but impossible and mixed signal techniques are required to overcome the problems of designing and analogue circuit on an essentially digital technology.

This unit will focus on CMOS Integrated Circuits. In the early nineties CMOS was used almost exclusively for logic circuits with Bipolar technology being used for the analogue functions as the latter offered higher frequency and greater power handling. The very high frequency specialised market was dominated by Bipolar on Gallium Arsenide. The fabrication capability in CMOS circuits has continuously increased since the early eighties and this has facilitated an ever-narrower channel in FET devices which, in turn, has led to increased speed. By 2000 the speed of CMOS circuits had matched that available from the exotic and expensive Bipolar on GaAs in 1994. Since then the reduction in channel length has exceeded expectation such that the anticipated 90nm channel length of 2007 was achieved in 2003 and the anticipated 60nm channel length of 2010 was achieved in 2007.

The heavy market dominance of digital CMOS technologies has necessitated analogue design on only very slightly modified digital processes. If a given analogue circuit can be realised on a very cheap process, then the cost per chip will put it in a very strong market position.

In 2007 80% of TSMC’s fabricated IC output had Mixed Signal (i.e. partly analogue) content. This was correctly predicted to rise to 90% by 2010. For an IC containing 10% analogue content, 50% of the design effort will be placed in this area and it will cause 70% of the post-prototype reviews.

1.2. The MOSFET

1.2.1. Device Physics

Before proceeding any further, it is essential that the basics of MOSFET operation are well understood, however, this is not a device physics unit. The reader is referred to Chapter 1.2 in [1], Chapter 2.1 in [2], and Chapters 4.1 and 4.2 in [3] for the required background device physics required for this unit.

1.2.2. Symbols

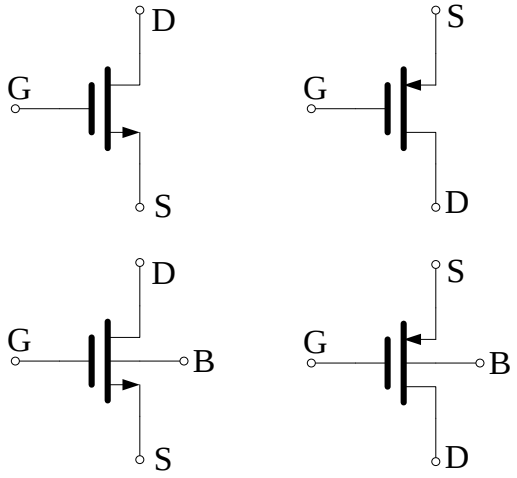


Figure 1: Common-use symbols for MOSFETS: N-channel (also known as NFET and N-type) on left and P-channel (also known as PFET and P-type) on right: Shown without Bulk Terminal (above) and with Bulk Terminal (Below).

Whether the mode of operation is ‘enhancement’ or ‘depletion’ is not explicit in these symbols. Most MOSFET devices in non-discrete form are enhancement mode.

1.2.3. Nomenclature

When examining the terminal characteristics of the FET the following nomenclature is used.

v_{DS} Voltage at the drain terminal with respect to the source terminal.

v_{GS} Voltage at the gate terminal with respect to the source terminal.

v_{GD} Voltage at the gate terminal with respect to the drain terminal. This is not commonly used in circuit design.

i_D Current **into** the drain terminal.

i_S Current **out of** the source terminal.

i_G The current into the gate terminal. This is zero in DC terms due to the extremely high gate resistance. In AC terms there is some flow of current due to the junction capacitance.

1.2.4. Terminal Characteristics: Large Signal Model

Fixing v_{GS} and varying v_{DS} for an enhancement mode MOSFET (the most common type available to the analogue designer on a digital-based process) will give one of the family of curves shown below in Figure 2. V_t (threshold voltage) is a process parameter and is the v_{GS} required in order to induce a channel between source and drain; increasing v_{GS} further enhances this channel. There are three distinct regions of operation:

- Linear region: $v_{DS} << v_{GS}-V_t$
- Triode region: $\sim V_t/10 < v_{DS} < v_{GS}-V_t$
- Saturation region: $v_{DS} > v_{GS}-V_t$

- The device is said to be ‘cutoff’ when $\boldsymbol{v_{GS} < V_t}$

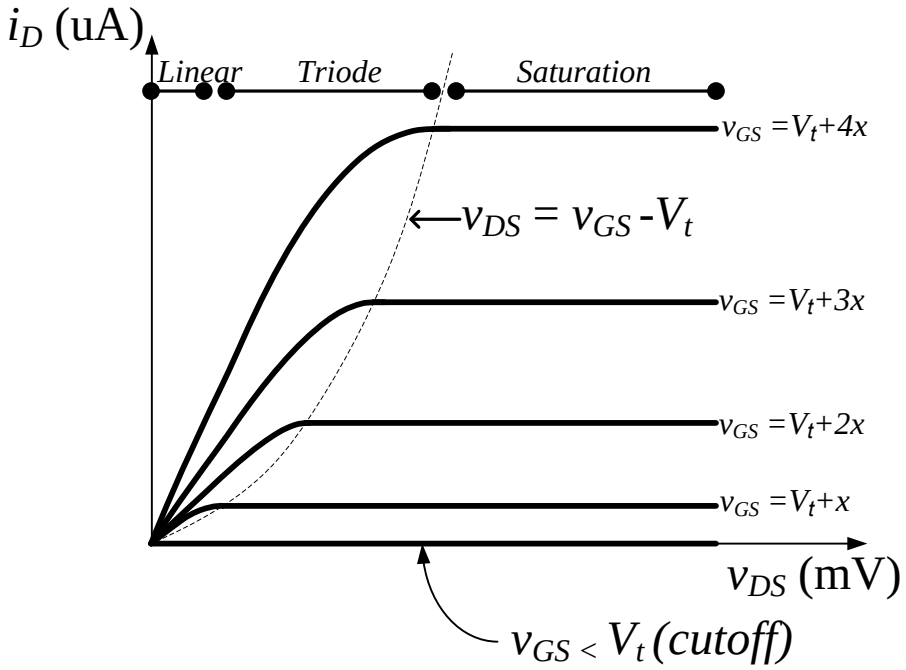


Figure 2: FET terminal characteristics for a representative device (scales are indicative only and may be considered as ‘magnitude’ to represent both N-channel and P-channel devices).

When employed as an analogue amplifier, the saturation region is used. In this region, as a first order approximation, $\boldsymbol{i_D}$ is independent of $\boldsymbol{v_{DS}}$ and varies predominantly as to the square of the overdrive voltage (the amount by which $\boldsymbol{v_{GS}}$ is larger than $\boldsymbol{V_t}$), see Figure 3; below. Note that ‘exponential’ and ‘velocity saturation’ (approaching linear) regions of operation are also identified.

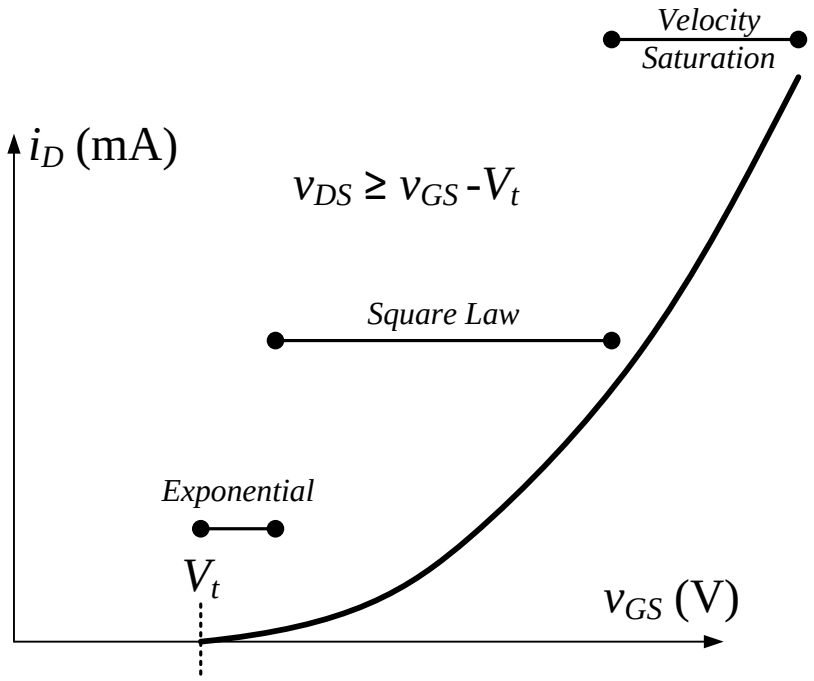


Figure 3: i_D-v_{GS} characteristic for an enhancement MOSFET in saturation (scales are indicative only and may be considered as 'magnitude' to represent both N-channel and P-channel devices).

For hand calculations the following approximations to the true terminal characteristics of the FET device are commonly used.

In the triode region i_D is given by

$$i_D = k\left[V_{OV} - \frac{v_{DS}}{2}\right]v_{DS}$$

Where V_{OV} is the overdrive voltage (v_{GS}-V_t), and k is a constant made up of several process parameters, the dimensions of the device (Width and Length) and physical constants:

$$k = k'\left(\frac{W}{L}\right) \qquad k' = \frac{\mu\varepsilon_0\varepsilon_r}{t_{ox}} = \mu C_{ox}$$

In the saturation region i_D is given by

$$i_D = \frac{k}{2}(V_{OV})^2$$

1.2.5. Small Signal Model: Transconductance and Output Resistance

In small signal terms, the transconductance (small signal change in output current to input voltage) is given by the slope of the i_D-v_{GS} characteristic at the operating point. There are three inter-related expressions for the transconductance (g_m).

$$g_m = k(V_{OV}) = 2\sqrt{ki_D} = \frac{2i_D}{V_{OV}}$$

There are three relationships for determining g_m, and there are three design parameters, (**W**/**L**), V_{OV} and i_D, any two of which may be chosen independently. It is common in contemporary IC design at gate lengths of greater than (say) 200nm to set the overdrive voltage to be 0.2V. The reason for this will be demonstrated in the laboratory material.

In the large signal terminal characteristics, as an approximation, gate length modulation is not included: Section 4.2.3 in [3] explains this phenomenon well. In summary, it should be understood that there is some dependence of i_D on v_{DS} due to the electrical field moving the channel pinch-off point away from the drain. In small signal terms, for a device in saturation, this may be adequately modelled by a linear resistance across the drain and source of the device. This resistance, r_{DS} or r_o, is given by

$$r_{DS} = r_o = \frac{V_E L}{I_{DS}}$$

Where I_{DS} is the large-signal DC approximation to i_D and the process parameter V_E is generally 5V/μmL for N-channel FETs and 8V/μmL for P-channel FETs. Although similar in terms of terminal characteristics, this parameter is not analogous to the Early Voltage in bipolar devices as different physical mechanisms are quantified: for FETs, V_A is used as an 'Early-type' parameter that is equal to $V_E.L$. James Early's work was with BJTs.

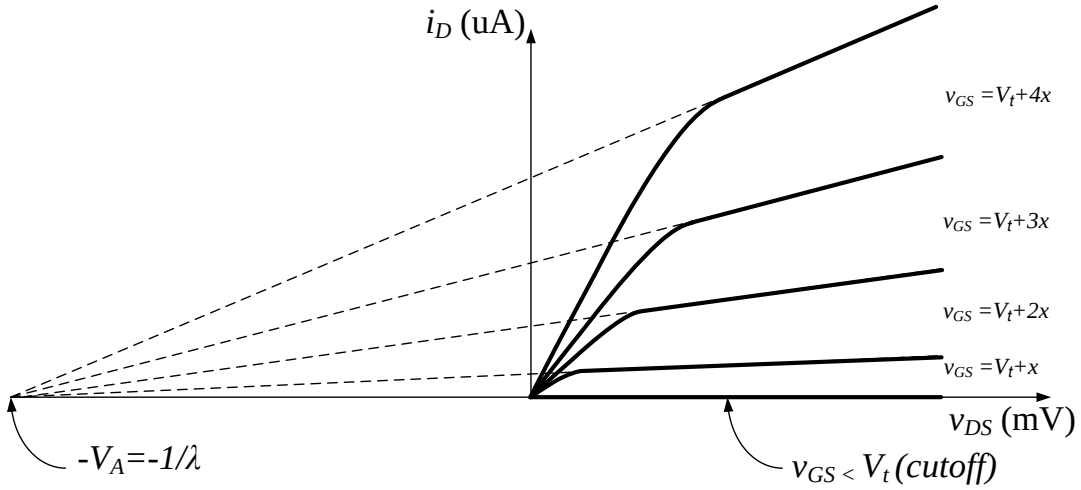


Figure 4: Affect of v_{DS} on i_D in the saturation region.

The parameter λ is commonly used as the channel length modulation parameter: $\lambda = \frac{1}{V_E L}$.

Making $r_{DS} = \frac{1}{I_{DS}\lambda}$.

2. CMOS Fabrication Process

Silicon (**Si**) is by far the most common material from which integrated circuits are realised, although Silicon-Germanium (**SiGe**) and Gallium Arsenide (**GaAs**) are also used. The basic fabrication steps are summarised below; there is an excellent summary of the process in Appendix A of [3]. The reader is also referred to Chapter 2.1 in [1], and Chapter 2.3 in [2].

The process starts with a CMP processed 400um-600um thick wafer of single-crystal Silicon. CMP is 'Chemical-Mechanical Polishing' (also known as Chemical-Mechanical Planarisation), a process that brings the surface of the wafer to a mirror-like finish. The wafer is usually between 10 and 30 cm in diameter. The conductivity of silicon is a function of the density of the impurities that it contains and of temperature.

The operations carried out on the wafer in order to realise an IC are:

- Oxidation: forming (growing) a layer of **$Si0_2$** on the surface of the wafer by heating it in an oxygen-rich environment. Silicon dioxide is an insulator with a dielectric constant of 3.9.
- Diffusion: Usually at high temperatures, impurity atoms (dopants) enter the solid Silicon from a dopant-rich environment thus changing its conductivity. The most common dopants are Boron, which will form a ***p*-type** region, and Phosphorous & Arsenic which both form an ***n*-type** region. The regions of diffusion on a waver are defined by an oxide mask pattern.
- Ion Implantation: a high-energy beam of ions are launched at the wafer as an alternative (low temperature) method of adding dopants to the silicon.
- Chemical-vapour Deposition (CVD): Pluralities of gasses are reacted with the wafer as a physical catalyst. The resulting material is deposited on the surface of the wafer. **$Si0_2$** and **SiN_4** (an alternative insulator) may be deposited by this technique. Also, epitaxial silicon may be grown on a crystalline silicon substrate (the single crystal lattice is maintained), and polycrystalline silicon (polysilicon) may be deposited on an oxide layer. The latter is commonly doped heavily to form local conductive interconnects and transistor gate connections.
- Metallisation: The wafer is coated with a layer of metal (aluminium, copper or gold) usually via a 'sputtering' process. The required pattern for component interconnection is then formed by etching away unwanted areas.

The patterns left by all of these operations are defined by a photolithography process in which a patterned photoresist layer masks the material below. This is used in two ways: the photoresist pattern is used itself as a block for (normally shallow) dopant implantation in to the silicon, or an oxide layer is pattern-etched and the residual areas mask the silicon below from (normally deep) implantation. Wet etching or reactive ion etching may also be used.

A CMOS fabrication process (known as a 'technology') is identified as ***n-well***, ***p-well*** or ***twin-well***. Currently, ***n-well*** is the most popular and this is introduced here (there are also ***triple-well*** processes.

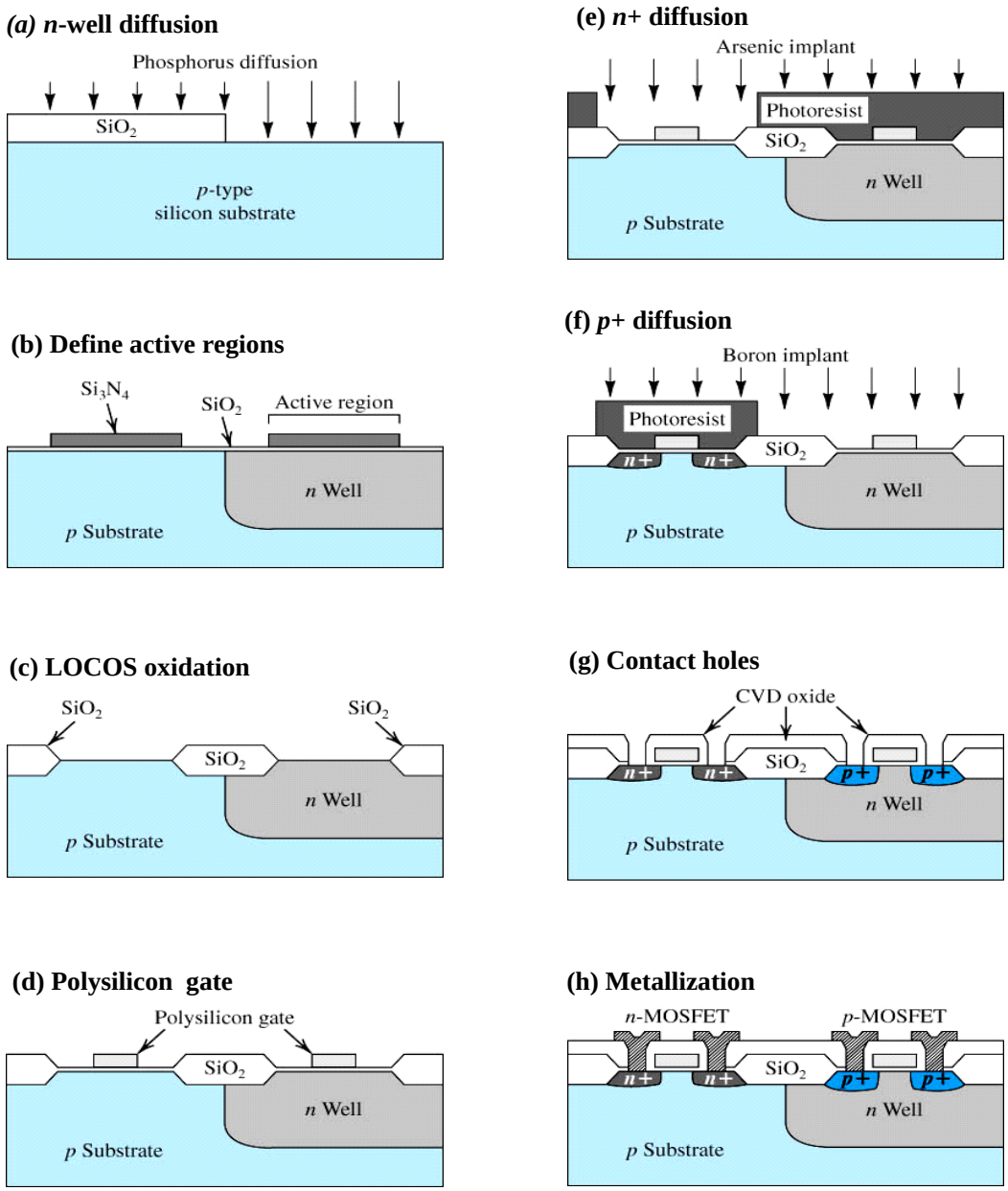


Figure 5: CMOS fabrication steps [3].

(a) A *p*-type substrate (usually epitaxial) is patterned with oxide via a photolithography process. The exposed substrate is doped with an ***n*-type** impurity to form ***n*-wells**. Ultimately the PFETs will reside in the ***n*-well**.

(b) After a thin 'pad oxide' layer is laid on the wafer to protect the lattice from the stress induced by the thick oxide growth, thin areas of nitride (i.e. deposited rather than grown) define where the active regions will be (devices).

(c) 'Thick-field' oxide is grown in the regions not protected by the nitride block. The process is called LOCOS (LOcal Oxidation of Silicon). The oxide layer is then etched back to reveal what will become the active areas.

(d) High quality, very thin (e.g. 20Å) oxide layer is deposited to form the gate oxide and a heavily (***n*-type**) doped polysilicon layer is deposited and patterned to

form the transistor gates. Some local interconnectivity may also be realised in the polysilicon layer.

(e) and (f) Arsenic (for ***n+*** contact implants) and then Boron (for ***p+*** contact implants) is diffused into the wafer for the Source (S) and Drain (D) regions. This is only a shallow infusion and so the photoresist layer is sufficient as a mask. If an oxide layer were to be required then its subsequent removal would break down the very thin gate oxide. Note that the diffusion passes through the thin gate oxide that is over the S+D areas and that the polysilicon gate acts to inhibit implantation in the channel regions.

(g) A thick layer of oxide is deposited and pattern etched to open contact holes to the S+D silicon areas.

(h) A metal layer is sputtered onto the wafer and etched to form the interconnect pattern of the first metal layer. Additional metal layers are added by successive iterations of stages (g) and (h).

Additional processing stages are used in a Bi-CMOS process in order to realise a ***npn*** vertical bipolar transistor. Having this high transconductance device facilitates accurate voltage (and current) references and higher speed circuit performance.

In a full-custom layout the designer will lay polygons in a graphical layout tool that describe the masks for the layers included in the process.

The process used in this unit does not include a Salicide layer.

Resistors and capacitors are also realisable using this process but will not be available in a digital-only design kit. Adding these components alongside the provision of better analogue models (including statistical parameter variance) is usually termed a Mixed Mode Design Kit.

Two forms of resistor are common. One is formed by implanting a length of n-well in the P-substrate and forming ***p+*** contacts at each end. The resistance is defined by the resistivity of the implant and its physical dimensions. The other common type is a deposited length of polysilicon on field oxide; the same resistance-dependence applies.

Two forms of capacitor are common. One is formed by the gate/thin oxide/channel structure in a relatively large FET. This capacitor has disadvantages of nonlinearity as the capacitance is dependent on the applied DC voltage to the gate and is also beset with the parasitic capacitances associated with the FET structure. A capacitor formed between metal layers (MIM-Cap, Metal-Insulator-Metal Capacitor) is generally preferred for accurate device construction. Here an additional metal layer is used between two of the standard process layers that is very near to one of the two. By this measure the oxide between the plates is thin and so greater capacitance per unit area is achieved. Vias are used to connect the standard higher metal layer to the additional inter-layer metallisation.

3. Products and Projects Flow

3.1.Microelectronics Product Development Lifecycle

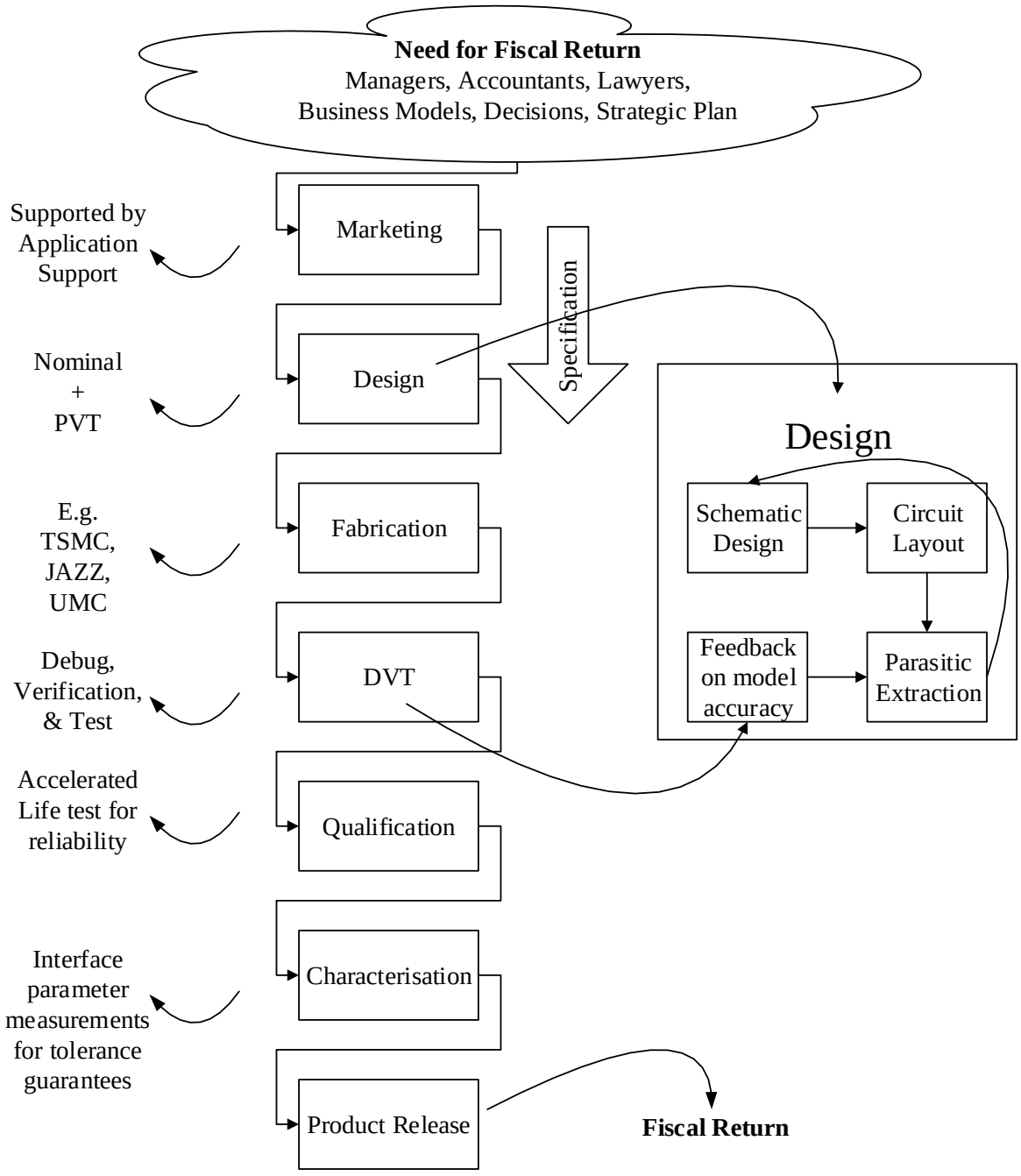


Figure 6: Microelectronics Product Development Lifecycle

3.2.Microelectronics Project Flow

From a project standpoint there are two drivers; Technology Capability and System Definition. Less complicated projects may be driven by either one of these. Starting with a System Definition is a 'Top Down' flow and starting with a Technology Capability is a 'Bottom Up' flow.

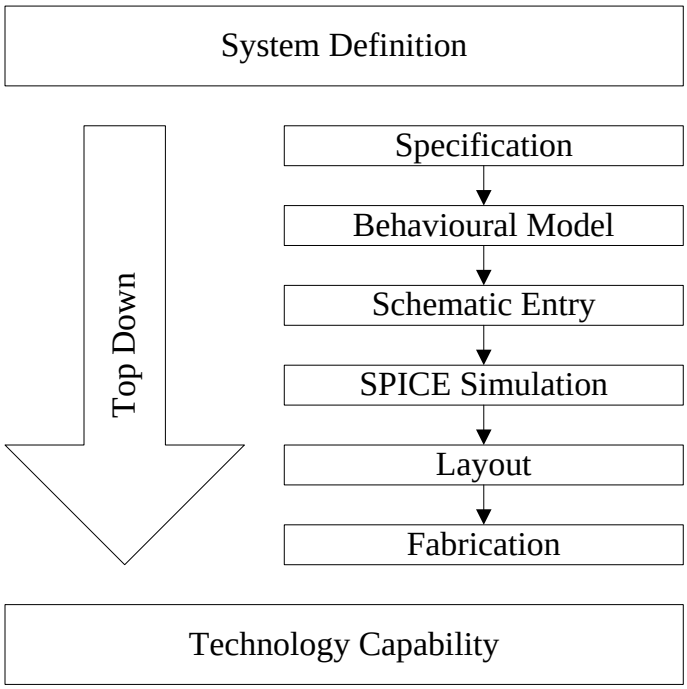


Figure 7: Top Down Flow for Analogue Integrated Circuits.

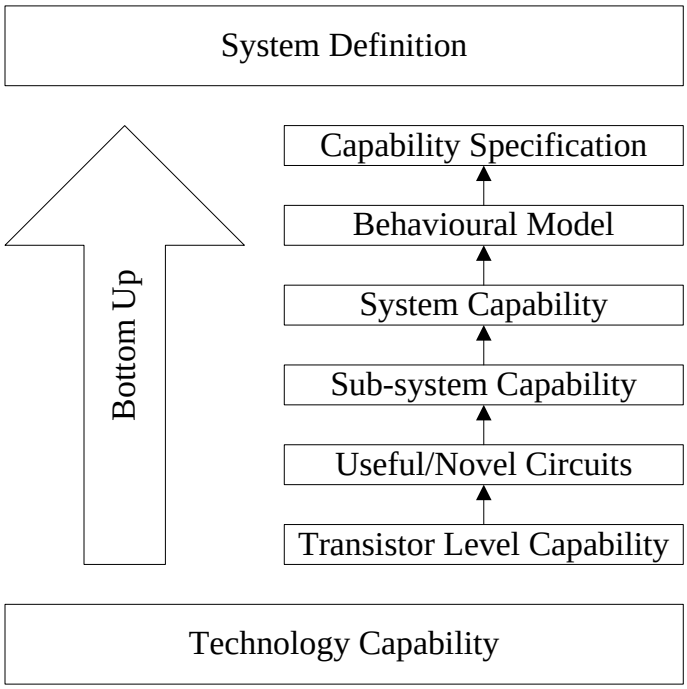


Figure 8: Bottom Up Flow for Analogue Integrated Circuits.

Both approaches have their limitations. The Top Down approach often neglects new innovations in fabrication capability and all the design work is left until the end of the process. The Bottom Up approach is obviously inappropriate for microelectronics design where a great deal of system-level control is required in order to manage a project.

Standard practice in analogue (and indeed mixed signal) integrated circuit designs is to use a Top Down Bottom Up project flow (TDBU). In general, the majority of the flow is TD but a specific team is identified within the project to address the lower half of the

BU remit. It then takes careful management to make the two flows meet appropriately in the middle.

3.3.IC Design Flow for Analogue Circuits

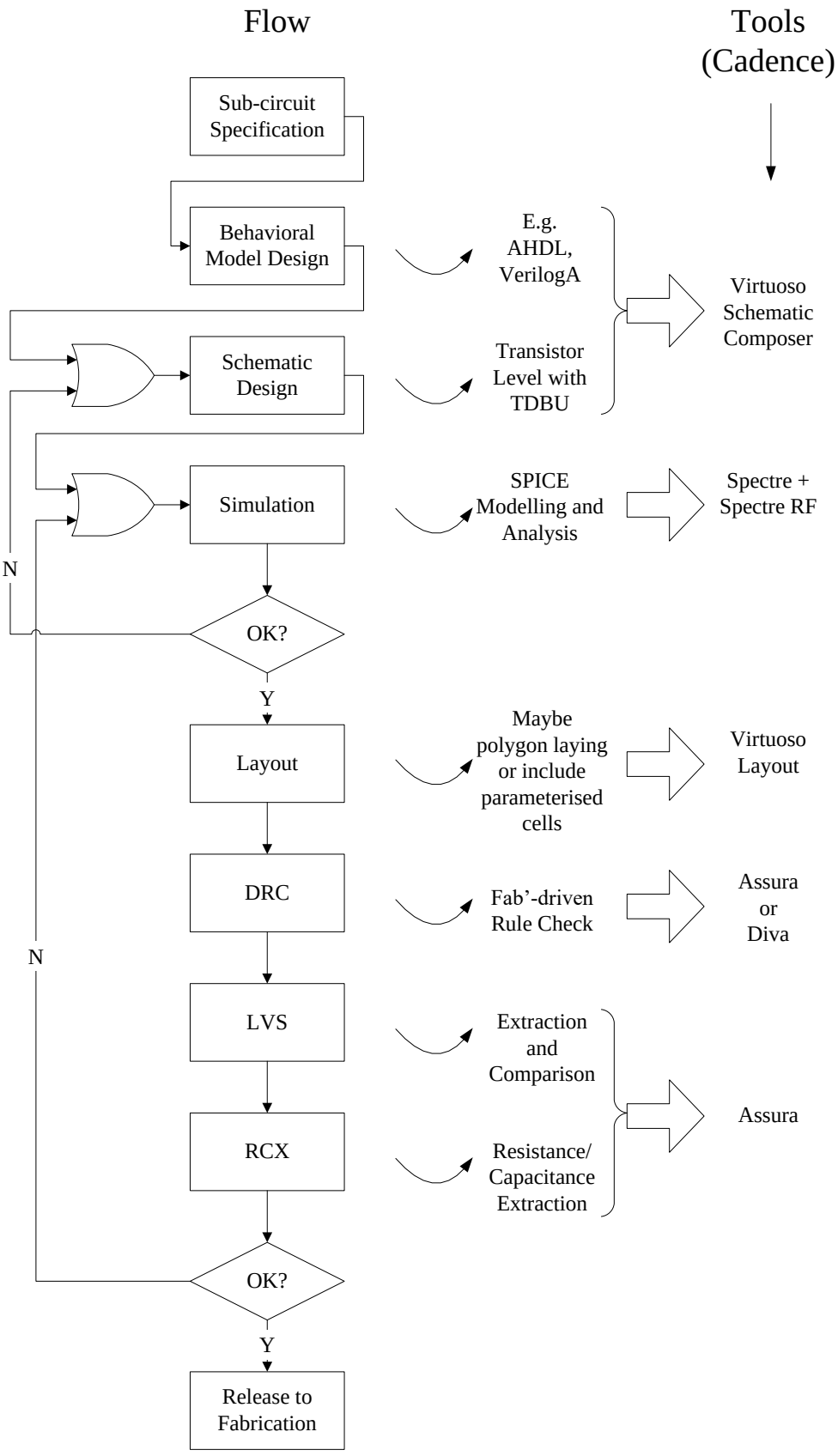


Figure 9: The Flow as would be carried out for each sub-circuit in a design.

3.3.1. Sub-circuit Specification

The primary specification of a sub-circuit is in the form of interface signal parameters (levels, noise, Bandwidth, CM (common mode level), $I_{ref,}$ etc.) and functional requirements (algorithmic). There may well be secondary specifications such as power budget, layout area/shape, CMRR (common mode rejection ratio) and PSRR (power supply rejection ratio).

3.3.2. Behavioural Model Design

In the design of analogue integrated sub-circuits, modelling is best done at the schematic level but with behavioural rather than device/component entities in each operational block. By this method, the functionality of the solution on an algorithmic level can be established in isolation from the circuit and component geometry design. The emphasis here is on *Schematic-driven design*.

3.3.3. Schematic Design

This is the core function of an analogue circuit designer. Each operational block identified at the behavioural level is enacted with an analogue circuit. The general topologies are known to experienced engineers and the component geometries are defined by a mixture of understanding, design rules and iterative simulation. Schematic design is explored in detail in Section 5.

3.3.4. Simulation

Even an experienced analogue designer is only expected to be aware of the linear and quadratic equations describing the terminal characteristics of a FET, whereas the SPICE models in use for IC design (or alternatively BSIM and EKV models) offer greatly enhanced accuracy at the expense of complexity. Therefore, the design process is somewhat iterative even when using nominal component values. Various forms of simulation are employed and are either directly based on SPICE simulation or on extension algorithms for periodic stimuli, stability and noise analysis. Simulation is explored in detail in Section 9.

3.3.5. Layout

Here, the masks for the different production layers used in the fabrication process are defined. This can be a completely manual process (known as ‘polygon pushing’) or some degree of assistance/automation may be employed such as connectivity awareness and/or Parameterised Cells. The former of these affords the designer ‘flight lines’ that show how components are connected in the schematic and the latter offers multi-layer cells that are derived from the components in the schematic. Layout is addressed in Section 10.

3.3.6. Design Rule Check (DRC)

A particular process will have a rule-set defining the proximity of diffusions, contacts, poly-silicon, metals, MIMCAP borders etc. Checking the design is best left to the automated process. Achieving ‘DRC-clean’ design is a landmark in the layout process. In addition to the DRC, and usually at the same time, an Electrical Rule Check (ERC) may be carried out. This algorithm checks for current density, antenna and a few other specialised rules such as electromigration.

3.3.7. Layout Vs Schematic (LVS)

Once laid out, the circuit conformity (both connectivity and component geometry) between the layout view and the schematic view must be ensured. This is done via an extraction process that generates a netlist from the schematic view and a second from the layout view; the two netlists are then automatically compared and a difference report produced.

3.3.8. Resistor Capacitor Parasitic Extraction (RCX)

Thus far in the design, the intentional circuit parameters have been considered whilst parasitic components have been ignored. Each network (net) has been considered as an ideal short circuit between all the components it connects. However, this is not the case in the layout realisation of the circuit. Each net will have a distributed resistance and a capacitance to the substrate and to nearby other conductors/features. Now that the geometry for the net is known, its parasitic elements may be automatically calculated and back-annotated onto the schematic. In fact the schematic view is rarely rewritten and ideal distributed parasitic elements are added to the layout view to form an ‘extracted layout view’(rc). In some cases the inter-network capacitance is also calculated resulting in an rcc view, however, this is commonly computationally intensive.

4. Analogue Design Trade-offs

There are many considerations that the analogue designer should be aware of throughout the design process and none of them should be left to the end. There is no substitute for practice and experience in establishing which consideration is the most important for a particular design but Figure 10 shows the trade-offs that will be encountered in most designs.

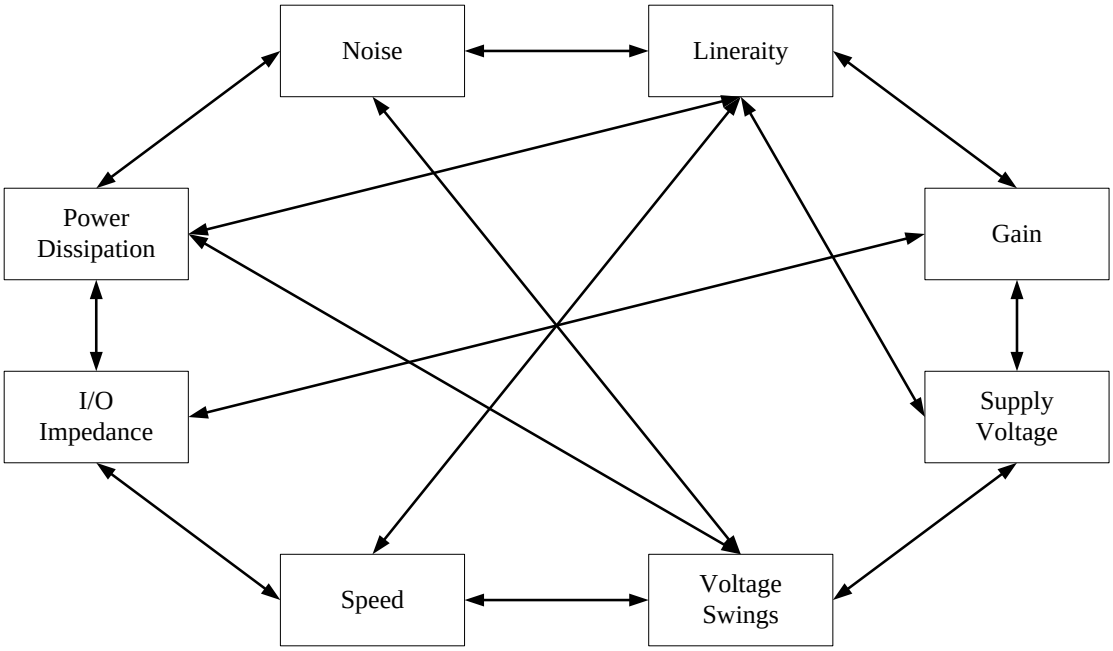


Figure 10: Analogue Design Trade-off Summary

5. Schematic Design

5.1. Designing an Analogue Circuit

The design of an analogue CMOS circuit to meet a specified functionality may be split into two parts.

- Schematic Topology

Here the device interconnects are defined such that a function is realised. For example, a differential voltage amplifier may be constructed from a pair of transconductance stages that share an AC ground and individual load stages that convert the output current into a voltage. This topology is biased via a current mirror and may be followed by a voltage buffer. The separate sub-circuits are placed together to form a circuit functionality.

- Component Sizing

Here circuit considerations such as required gain, output voltage swing, input common mode range and die area are used to define the geometries of the active and passive devices alongside the bias current and overdrive voltage.

This unit focuses mainly on the design of suitable schematic topologies and the resulting performance is taken as acceptable from (in the most part) equal-sized FET devices of 10μm wide by 1μm long. Component sizing is explored in detail for an

example circuit: the two-stage operational amplifier. In this circuit, stability issues are important making it a worthy example to explore.

5.2. Schematic Entry

You will be familiar with the process of constructing a circuit schematic for SPICE simulation from previous study and/or the VLSI3 unit taken as part of the AMSE programme. One useful additional construction technique is the use of busses on analogue (or digital) nodes which enables much tidier graphic presentation when defining multiple instantiations of the same component or signal line. This technique is summarised in Figure 11.

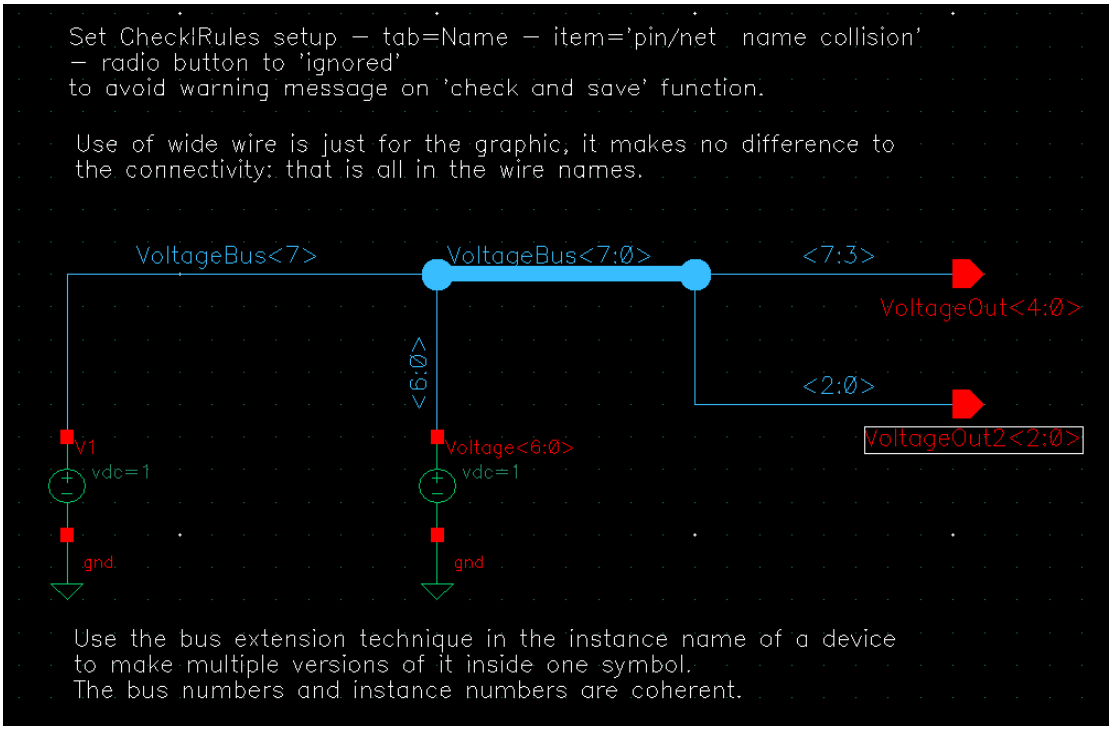


Figure 11: Use of Busses in Schematic Entry

5.3. Libraries for Analogue Design

This section introduces the component libraries in the design software used in this unit that are suitable for analogue design.

5.3.1. analogLib

The *analogLib* library contains useful ideal elements for initial circuit design; active, passive, dependent and independent. This is useful for exploring circuit theory and testing the realisation of functions within an ideal topology.

5.3.2. basic

The *basic* library contains fundamental building blocks that are used to generate a schematic such as pins, element symbols and defining elements such as *noConn* (an unconnected network notifier).

5.3.3. US_8ths

This library contains sheet borders and title blocks as should be found on all engineering drawings for configuration control. Schematic Diagrams fall within this category of drawings.

5.3.4. cdsDefTecLib

This is an unused library that is the default configured technology library. It is normally empty, but if it does contain element symbols, they have no functional model associated with them (neither ideal or SPICE).

5.3.5. gdpk180

This is the technology-driven process data kit that contains cells, symbols and SPICE models for the fabrication process used in this unit. In this case, the prefix **g** is for 'generic'; meaning it is not realisable but is representative of a worldwide state-of-the-art.

Section 14 provides an exhaustive account of the contents of the Technology Library used in this unit.

5.4. Test Bed Design

5.4.1. Rational

A schematic view of a circuit is designed in response to a specification, i.e. it will have a function to perform and will interface with other schematic-driven functional blocks in a system design via a defined interface. A circuit within a system can only rarely be designed in isolation.

Before the design of a schematic that represents a functional circuit can begin, the interfaces and functionality must be defined; this will be in a Design Specification document. Initially, the circuit is designed assuming ideal interfaces and only after a number of interconnecting circuits are designed may the statistical errors on the interfaces be included in the simulation. This approach means that the overall functionality of a system cannot be ascertained until all its parts are complete; however, this does not comply with the (specification-driven) top-down approach to project flow. Further, many systems are far too large to be simulated without a high degree of abstraction, meaning that most of its parts will be modelled by ideal *views* in order to test a small part of the system modelled by the SPICE *view*.

The practical way to overcome this limitation is to assume a worse-case error on the signals at a circuit's interface and simulate the functionality of it at these extremes in addition to nominal values. Effectively, this adds further corners to the simulation field. Only at a late stage in the design process may the error bounds on the interface signals be established accurately and design review is required in order to account for any errors beyond that expected.

In order to test a circuit in isolation of the other circuits at its interface, a test bed is required. There is no point in beginning a design with anything else and often the design of the test bed will bring previously unconsidered points to light. The test bed will offer the circuit the power and input interface alongside an appropriate load at the output. There may well be a feedback loop in the test bed if the function of the circuit is iterative and, commonly, there is some form of ideal processing on any information signal output

in order to establish its validity. There are two approaches to this; offline processing of raw result files and live processing as the simulation progresses. Any information-bearing signals are constructed ideally in the test bed so that the expected output is known. In some instances there may be an ideally constructed parallel path in the test bed so that the output of the circuit may be compared with one of an ideal nature.

5.4.2. Example

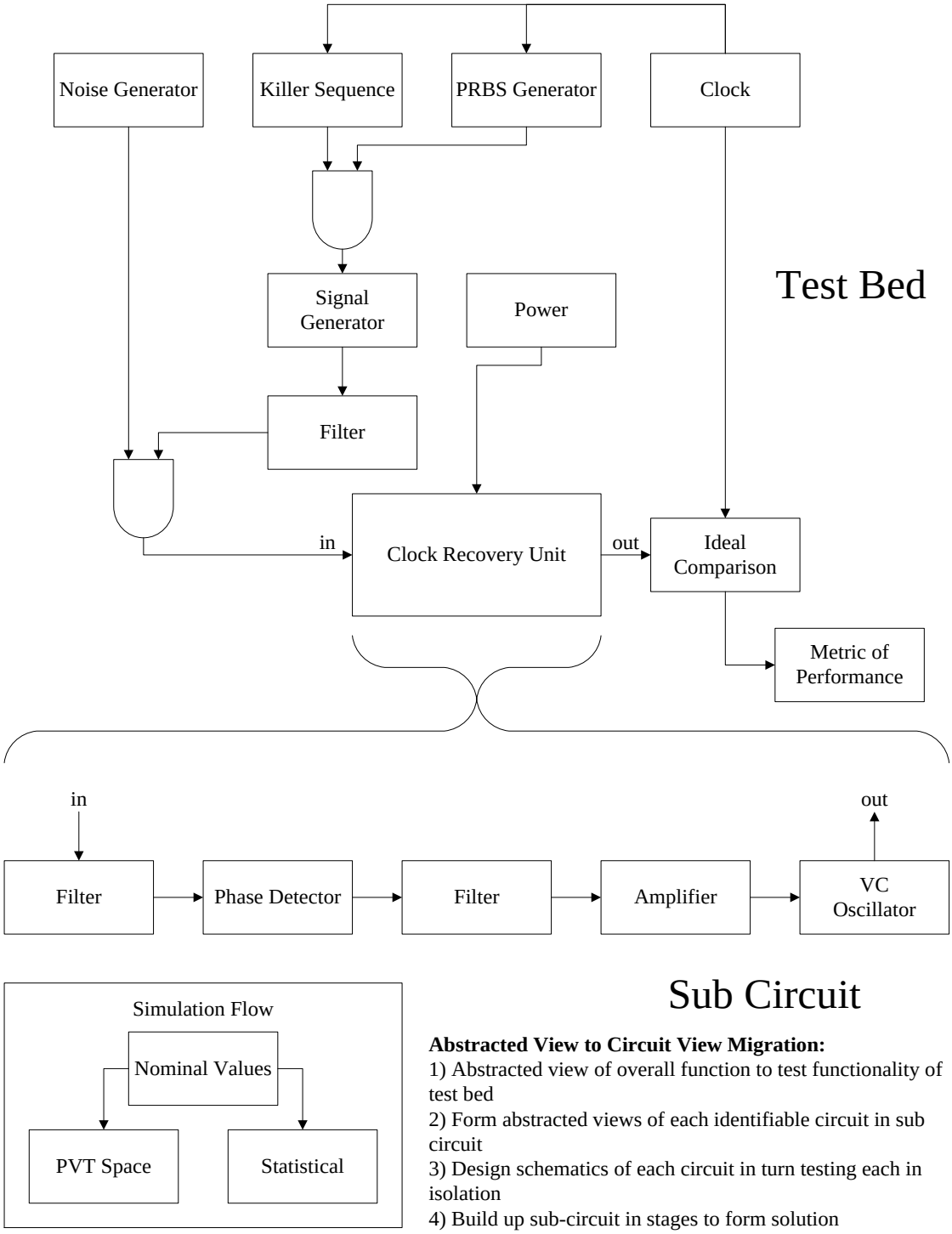


Figure 12: Example Test Bed Design

5.5.Schematic Format

5.5.1. Conventions

There are certain conventions that should be observed in the construction of a schematic diagram (electronic). A list of the most common is given below:

- Bias current flow is from the top to the bottom. Thus, the positive power rail should be at the top of the schematic and the earth rail should be at the bottom (or negative rail for balanced supplies).
- Signal flow (not necessarily current flow on the signal lines) is from left to right.
- Circuits are drawn so as to emphasise the operation on the information-bearing signal. The bias, CMFB etc. circuitry is included as a sub-circuit and its interconnection with the main circuit realised by net names rather than lines.
- Active and passive elements that are not part of a feedback path or an interconnect between stages are aligned top to bottom.
- Networks are drawn from horizontal or vertical sections except where this will make the functionality of the circuit awkward to ascertain (e.g. a cross connection in a balanced circuit).
- Balanced stages are drawn with components opposite one another.
- Only ‘solder dots’ (a small circle on two crossing wires) imply connection i.e. wires that cross without this feature are not connected. T-sections are assumed to be connections.
- A schematic should not be too large. If it contains too many components then its functionality is harder to ascertain. Hierarchy should be used i.e. sub-circuits are represented by symbols with connection pins. As a guide, 25-30 devices is about the top limit and one should aim to present a schematic with less than this.

5.5.2. Example

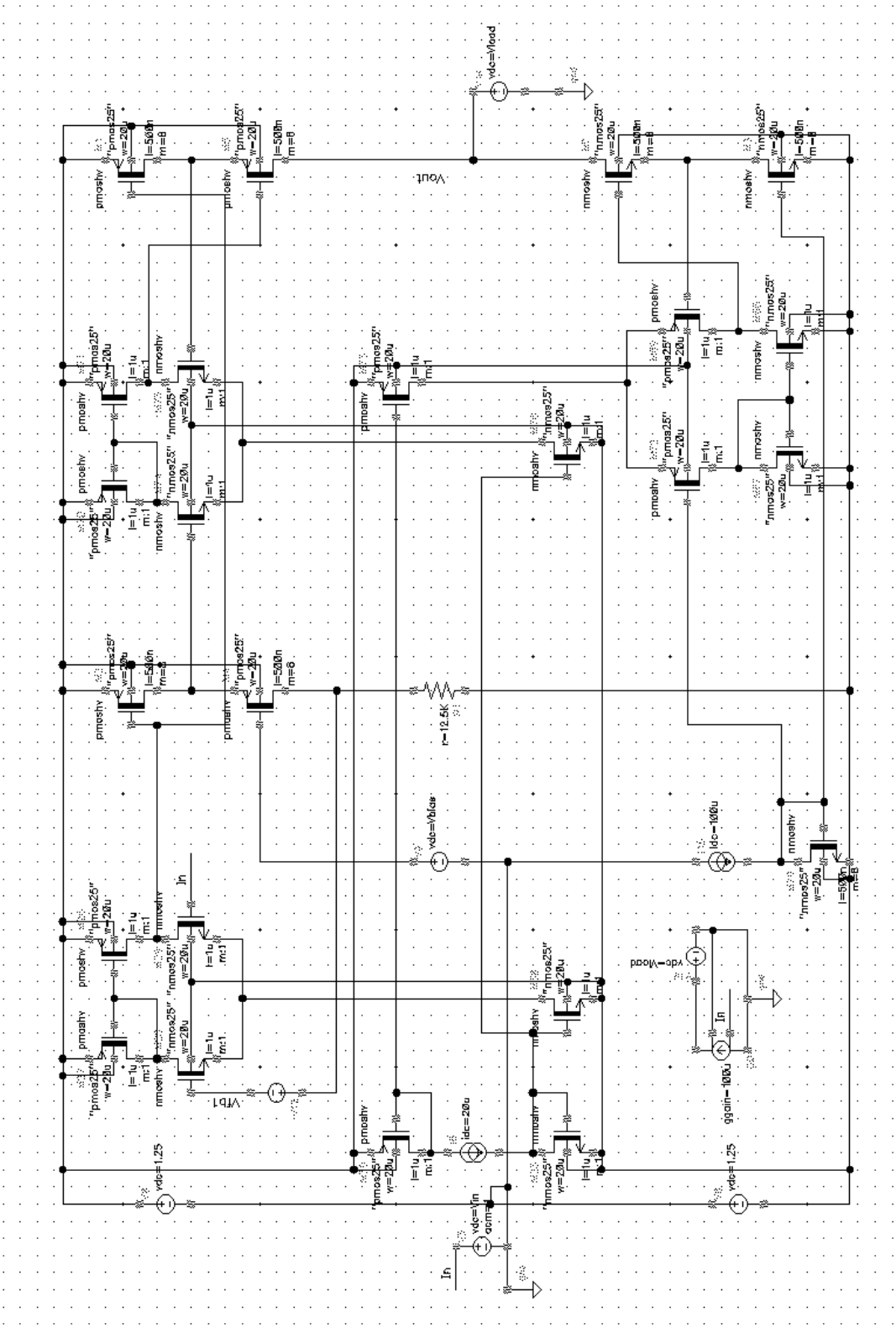


Figure 13: Example Schematic (to be considered in landscape presentation)

6. The Circuits used in the laboratory activities

6.1. Terminal Characteristics Evaluation

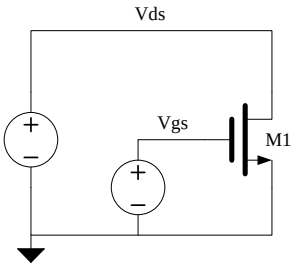


Figure 14: Circuit for Large Signal Transconductance Terminal Characteristics Evaluation

This circuit establishes the biasing conditions for an evaluation of the device terminal characteristics of transistor **M1**. V_{ds} is held constant and V_{gs} is varied as i_D is observed. This circuit is a basic transconductance amplifier (although here the method of measuring the output is not identified).

6.2. Voltage Amplifiers

6.2.1. Basic Voltage Amplifier

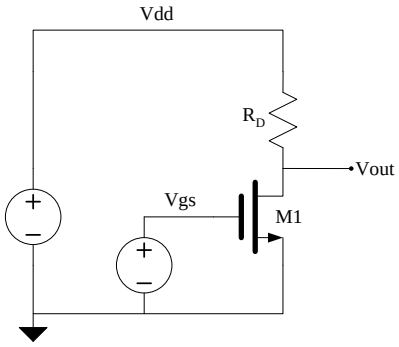


Figure 15: Resistive Loaded Voltage Amplifier Circuit

The output current drawn from V_{dd} generates a voltage at the output terminal due to the presence of a load resistor, R_D. As V_{gs} rises, i_D increases and so V_{out} falls from V_{dd}. This circuit is a basic (inverting) voltage amplifier with a gain determined by the product of the transconductance of **M1** (g_{mM1}) and the value of R_D. In small signal terms, the output resistance of **M1** should be taken into account as this is in parallel with R_D and so the equation for the small signal voltage gain of this circuit is given by

$$\frac{v_{out}}{v_{gs}} = G_v = -g_{mM1} \times (R_D \| r_{oM1}).$$

As r_{oM1} will be much larger than R_D, the former of these two controls the voltage gain of the circuit.

In terms of signals, one considers V_{gs} to comprise two parts, V_{GS} and v_{gs}. The former of these two is the steady state aspect of the parameter and the latter is the small signal aspect which varies in the time domain. Previously in these notes we have presented

the gate voltage in terms of v_{GS}, which is merely a large signal condition that is swept in the voltage domain (and hence has a lower case letter because it is not constant). In many cases V_{GS} is considered to be the bias applied to the device and v_{gs} is considered to be the signal superimposed on the bias.

The large signal transconductance, G_M, is the ratio of I_D to V_{GS} at the operating point.

The small signal transconductance parameter, g_m, is found by linearising the circuit at the operating point (due to the applied bias) and establishing the current-out to voltage-in rate-of-change.

$$g_m = \frac{\partial I_D}{\partial V_{GS}}.$$

6.2.2. IC Voltage Amplifier

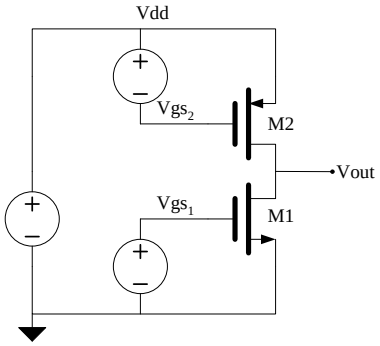


Figure 16: Current Source Loaded Voltage Amplifier Circuit

R_D in Figure 15 is replaced with a current source formed by **M2** under a constant gate bias V_{gs2} which may be identified as V_{GS2}. The internal resistance of the current source is comparable with that of the gain device and so, in small signal terms, the parallel combination of the two determines the translation of i_{DM1} to v_{out}.

$$\frac{v_{out}}{v_{gs_1}} = G_v = -g_{mM1} \times (r_{oM1} \| r_{oM2}).$$

This form of load is common in analogue IC design as greater resistance may be realised by current sources and, therefore, greater voltage gain is achieved. Should **M1** and **M2** be ideal (i.e. the gate length modulation is zero) then the voltage gain would be infinite.

The reader is referred to Section 3.2 in [1] and Section 5.1 in [2] for further information.

6.2.3. Cascoding for Greater Voltage gain

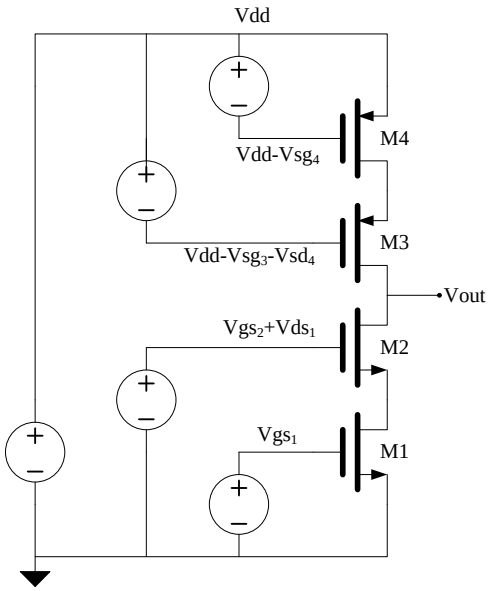


Figure 17: Cascoded TCA and Load Voltage Amplifier Circuit

In practical terms, a need for significant gain cannot be sufficed by increasing the output resistance of the load and gain devices alone as, for a given process this would result in a very long device with a very low bias current, $\boldsymbol{I_{DS}}$. The long device would take up too much die area and the low bias current would force the device into weak inversion and cause significant noise issues.

The output impedance of a FET gain stage may be increased by *cascoding*: following the *common source* device with a *common gate* device. An intuitive explanation as to why this happens is given here and the reader is referred to Section 3.6 (High Output Impedance Current Mirrors) in [1] for an analytical approach and derivation of the new output impedance at the drain of the common gate stage as being

$$r_o = g_{mMY} \times r_{oMY} \times r_{oMX}$$

where $\boldsymbol{r_{oMY}}$ and $\boldsymbol{r_{oMX}}$ are the output resistances of the common gate and common source devices respectively, and $\boldsymbol{g_{mMY}}$ is the transconductance of the common gate stage.

Consider the voltage at the drain of **M1** in the Current Source Loaded Voltage Amplifier circuit (Figure 16). It is the variation of this voltage that modulates the length of the channel of **M1**. If this voltage is fixed then no modulation occurs leading to the obviation of the need to consider the output resistance. Now consider the voltage at the drain of **M1** in the cascoded TCA and Load Voltage Amplifier Circuit (Figure 17). This voltage is fixed by the required overdrive voltage in **M2** and (the fixed) $\boldsymbol{V_{GSM2}}$ forming a low impedance node, whilst the drain voltage of **M2** is free to vary as it is a high impedance node. Effectively, one may consider that **M2** is buffering the drain of **M1** from the high impedance node so that only a small amount of voltage variation is observed at this point. The low voltage variation leads to a low level of channel length modulation for a given process and device geometry. There will be some voltage variation as the current through the **M1**-**M2** cascode changes. It is left to the reader to consider the meaning of 'low impedance node' in terms of how the circuit driving the node may be modelled as a voltage source.

6.3. Current Mirrors

Current mirrors are examined extensively in other units on this programme. The reader is referred to Section 4.4 in [2] for an analytical address of the common types. Here, an intuitive approach is used to explain the circuit operation for the basic and the cascode types.

6.3.1. Basic Current Mirror

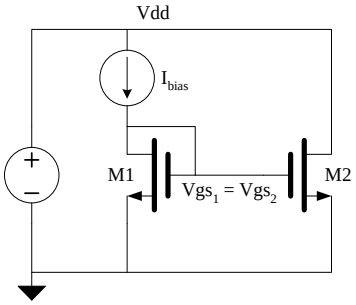


Figure 18: Current Mirror

Subject to the limitations stated below the circuit will draw a current equivalent to $\boldsymbol{I_{bias}}$ from $\boldsymbol{V_{dd}}$ through **M2**. As the gates of the two NFETs are connected they are under the same operational conditions and so will have the same drain currents. This assumes that they are exactly matched and have the same geometry. There is a mechanism inherent in the circuit for feedback control of the gate voltage although the output node is not included in this path. Consider the gate voltage as the circuit is enabled and $\boldsymbol{I_{bias}}$ begins to flow. Initially, there is no driver for the gate voltage and it will float. As $\boldsymbol{V_{dd}}$ and $\boldsymbol{I_{bias}}$ are established, the feedback from the gate of **M1** to its drain forces the voltage of this node to be that required for **M1** to pass $\boldsymbol{I_{bias}}$. **M1** will always be in saturation as $\boldsymbol{V_{gs}}$ and $\boldsymbol{V_{ds}}$ are the same. **M1** acts as a variable resistor under feedback control to establish an operating point such that the requirement to pass $\boldsymbol{I_{bias}}$ is satisfied. The presence of **M2** does not affect this operation as it is connected to **M1** via its high impedance gate. Indeed, **M2** may be repeated a number of times without changing the operation of the circuit. This is the basis of many current amplifier circuits.

Inaccuracy in the mirroring of current from **M1** to **M2** comes about via two mechanisms. The first is one of matching: process and temperature variation will cause an offset in the mirror. The second is one of finite output resistance (via channel length modulation): for the devices to mirror correctly, their $\boldsymbol{V_{ds}}$ values should be the same which is not the case here. **M1** has a $\boldsymbol{V_{ds}}$ equal to $\boldsymbol{V_{gs}}$ and **M2** has a $\boldsymbol{V_{ds}}$ of $\boldsymbol{V_{dd}}$. In some cases it is sufficient to overcome this issue by extending the length of **M1** and **M2**; however, this will raise $\boldsymbol{V_{gs}}$ and lower the output voltage range over which the circuit operates.

6.3.2. Cascode Current Mirror

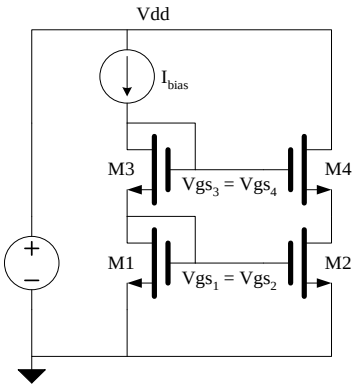


Figure 19: Cascode Current Mirror

The technique of cascoding may be used in the current mirror to increase its accuracy across a varying output voltage. Here, the simplest cascode approach is shown which is effectively two mirrors stacked on top of one another. This is a useful solution; however, it suffers from a minimum output voltage of (***2Vov+Vt***) in order to keep all devices in saturation. Alternative (and generally more complicated) cascode solutions aim to lower this restriction.

6.4.Balanced Output, Balanced Input Amplifiers

6.4.1. Differential and Common Mode Signal Components

A balanced signal is one in which the information content is carried in the difference in two *natures* of the system. This is true for all disciplines including magnetic, thermal and kinematic, however, here we limit our consideration to electrical systems in which the potential *nature* is Voltage and flow *nature* is Current. The balanced signal may be two currents or two voltages, however, the latter is more common unless high speed or high dynamic range is required in an analogue IC design. Here we consider a system in which two voltages form a balanced signal.

For the two signals v_1 and v_2, we define the *differential mode* signal as

$$v_D \equiv v_2 - v_1$$

and the *common mode* signal component as

$$v_{CM} \equiv \frac{v_1 + v_2}{2}.$$

It follows that $v_1 = v_{CM} - \frac{v_D}{2}$ and $v_2 = v_{CM} + \frac{v_D}{2}$.

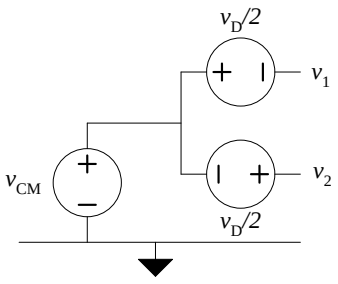


Figure 20: Signal Components

Balanced implementations have the distinct advantage over their ‘single-ended’ counterparts (in which the information is carried in the absolute value of a nature with respect to a system reference e.g. earth) that noise and interfering signals (including those on the power supply) is suppressed as this places a common mode disturbance on the pair which is ignored.

6.4.2. Basic Balanced Amplifier

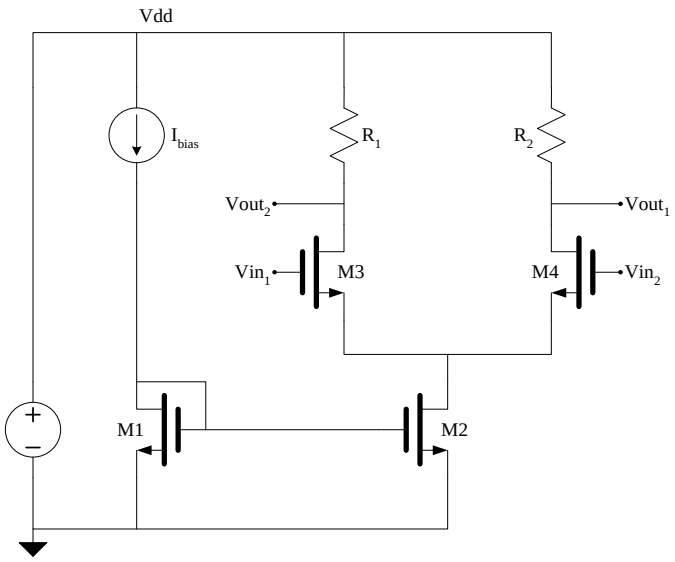


Figure 21: Resistor Loaded Balanced Amplifier

The balanced amplifier is a fundamental component in Analogue IC design comprising a current mirror and two voltage amplifiers. The current, $\boldsymbol{I_{bias}}$, is set, alongside the component geometries so that, for a zero differential input at the gates of **M3** and **M4**, $\boldsymbol{V_{out1}}$ and $\boldsymbol{V_{out2}}$ are near mid-rail voltage. This ensures a wide output voltage swing. The node at the drain of **M2** is free to follow the common mode component of the input pair and so $\boldsymbol{v_{CM}}$ is removed from the system as a driving signal.

An intuitive explanation for the operation of the circuit is given here and the reader is referred to Section 5.2 in [2] for a derivation of the differential mode gain.

In terms of the differential mode operation of the circuit, the node at the drain of **M2** is seen as a ground. This is because any incremental current driven via **M3** is balanced by a decremental current driven via **M4**, and visa-versa. This node is commonly known as an AC ground. If $\boldsymbol{V_{in1}}$ becomes positive with respect to $\boldsymbol{V_{in2}}$ then more current is drawn through $\boldsymbol{R_1}$ than $\boldsymbol{R_2}$ and so $\boldsymbol{V_{out1}}$ will rise whilst $\boldsymbol{V_{out2}}$ will fall (note the positions of $\boldsymbol{V_{in1}}$, $\boldsymbol{V_{in2}}$, $\boldsymbol{V_{out1}}$ and $\boldsymbol{V_{out2}}$ in the figure). Assuming $\boldsymbol{R_1}$, $\boldsymbol{R_2}$, and $\mathbf{M_{3,4}}$ are matched then the small signal voltage gain may be shown to be the same as a single transistor amplifier, i.e.

$$A_v = g_m R \qquad\qquad R = R_1 = R_2,\ g_m = g_{m3} = g_{m4}$$

The gain of this amplifier may be improved by the use of diode-connected loads, current source loads and cascoding as in the single-ended amplifiers. These circuits are shown below.

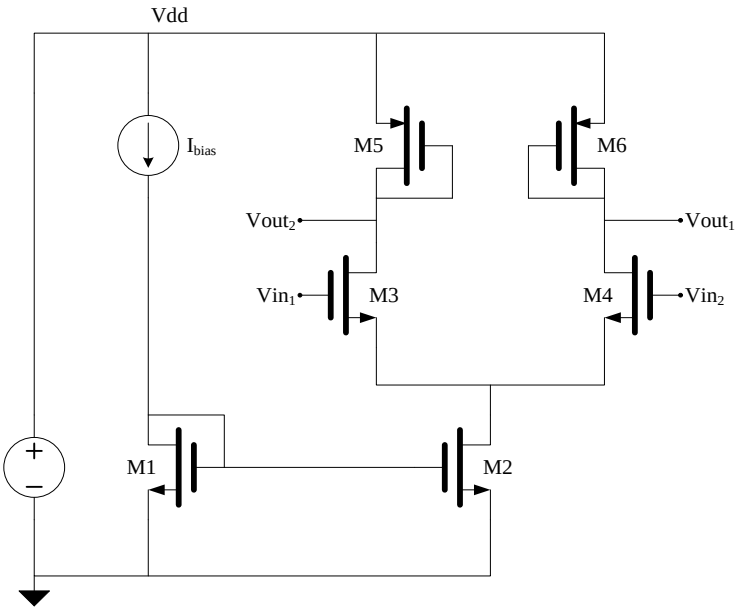


Figure 22: Diodo Loaded Balanced Amplifier

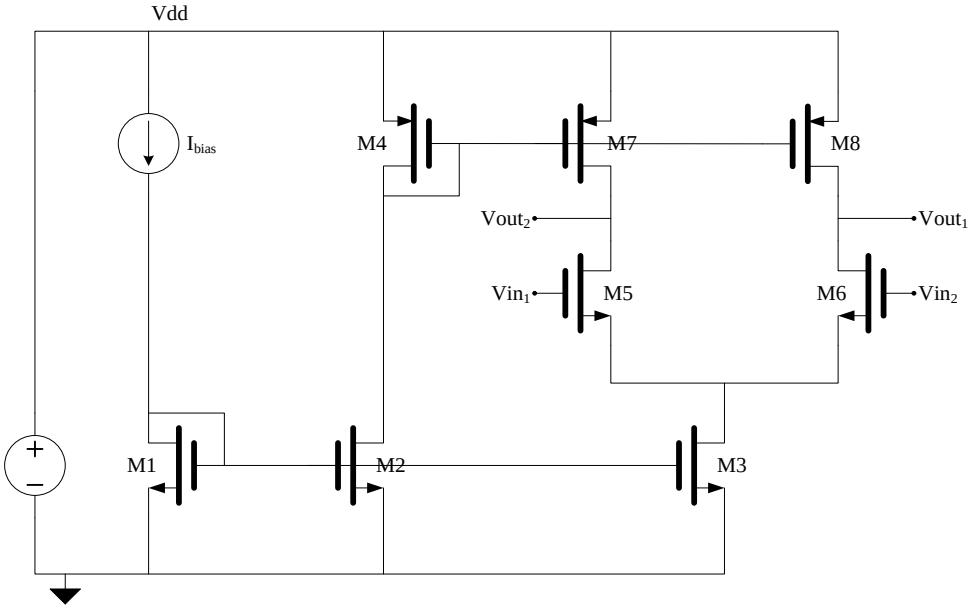


Figure 23: Current Source Loaded Balanced Amplifier

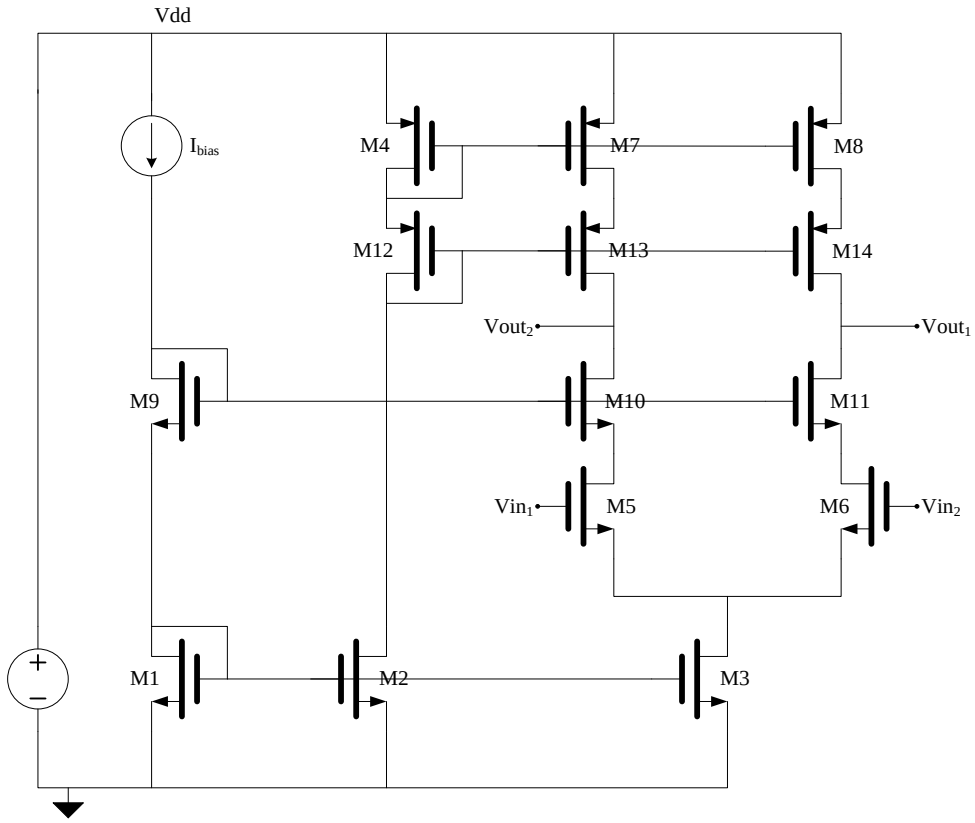


Figure 24: Current Source Loaded Cascoded Balanced Amplifier

When circuits become as populated with devices as that shown in Figure 24 it is common practice to keep the devices in pairs of columns where balanced signals are employed. Figure 25, below, shows this format where the tail current of the balanced amplifier is split across two devices that are themselves cascoded. It should be pointed out that, this is no longer appropriate for process nodes below (say) 250nm as the voltage headroom is insufficient for such cascade structures.

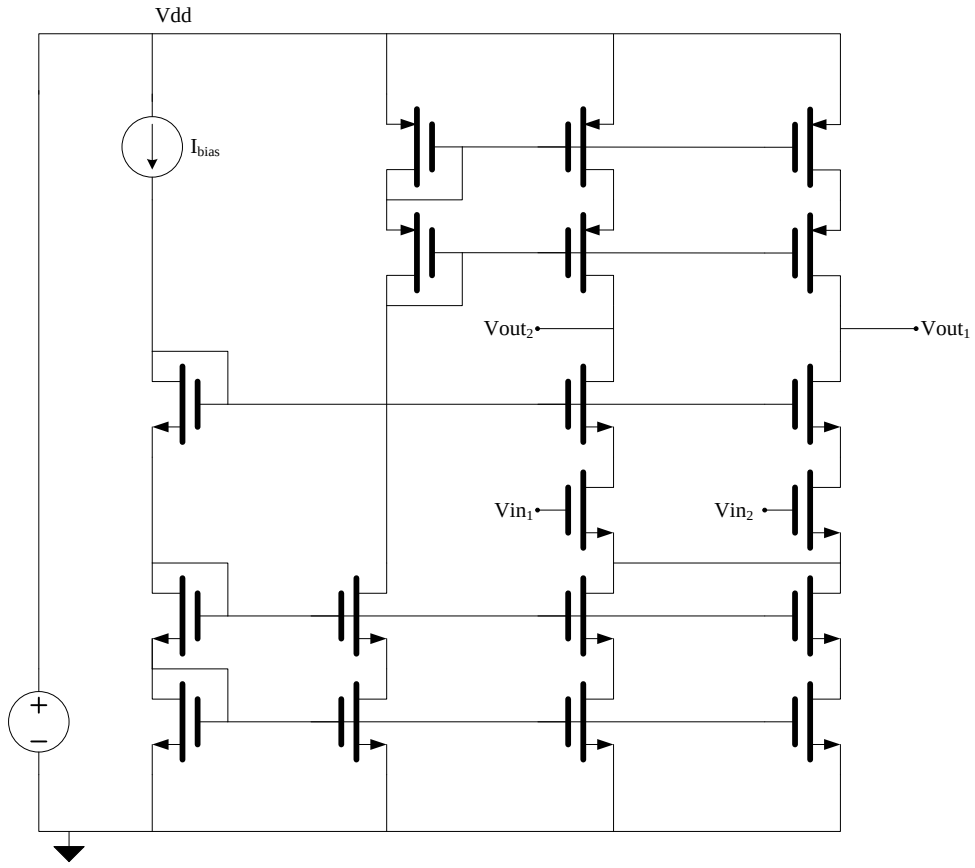


Figure 25: Current Source Loaded Cascoded Balanced Amplifier in Standard Form

6.5. Single End Output Differential Amplifiers and Operational Amplifiers

6.5.1. NFET Input Differential Amplifier

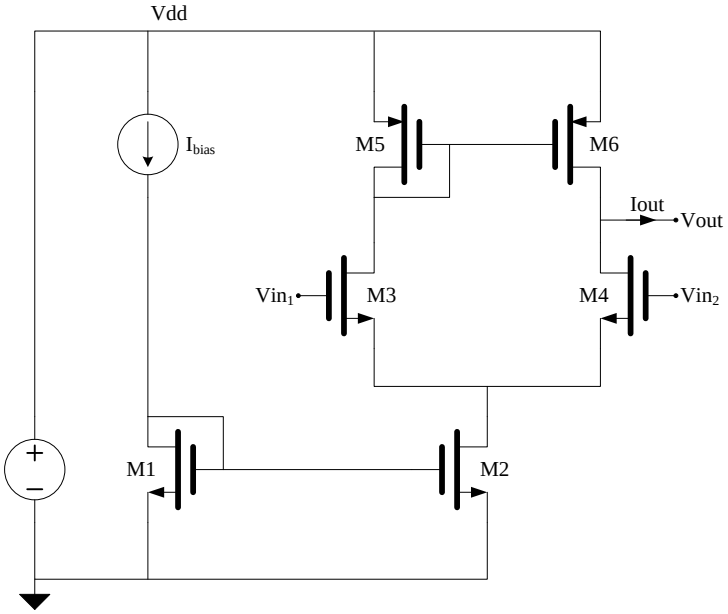


Figure 26: Current Mirror Loaded, NFET Input, Differential Transconductance Amplifier

A differential amplifier has a single output that is proportional to the differential mode of the two inputs. One could simply use only one output from the balanced amplifier configuration but this would not employ all the available gain from the topology. Here a current mirror load is employed. This gives high gain whilst maintaining the output voltage swing that would be lost if cascoding were used for this purpose. The current mirror load offers a dynamically varying current to the output node dependent on the input condition. As all devices are operating in current mode, and the output has a high driving resistance the output current is predominantly linearly related to the small signal differential voltage input, making this a TCA. Indeed, the output may be shorted to a mid-rail voltage source.

Section 5.2 in [2] gives an in-depth analysis and an intuitive explanation of circuit action is given here.

Should this circuit drive a low impedance load, it may be considered to offer a pure current output. However, in the high impedance load condition, little variation in the output current will be observed and the output voltage will change to satisfy circuit conditions.

The bias circuit and input devices are the same as the balanced amplifier, however, now the current mirror load acts to enhance the gain by dynamically altering the current drive of **M6** to oppose the current draw of **M4**. In comparison to the constant current source load implementation (or, alternatively a fixed resistor load) it may be considered that there are two mechanisms that drive the output voltage (if loaded with a high impedance or current (if loaded with a low impedance). The first being the standard action of the **M4** transconductance stage driving the high impedance node at the drain of **M6** and the second being the mirrored action of the **M3** transconductance driving the same high impedance node.

An alternative yet equally valid approach to understanding the operation of this circuit is to consider **M6** as a current source that opposes the demands of **M4**, this requires $\boldsymbol{V_{out}}$ to shift a greater amount in order to maintain compliance with the devices' terminal characteristics for a given differential input (high impedance load).

A further advantage of this circuit is not immediately apparent. A high gain is achieved without the need for additional circuitry to control the bias of the load devices. This would normally be required in order to approach this amount of gain which is far in excess of separate diode-connected loads. The shortcoming of the circuit is that it is limited to single-ended output operation.

The question remains of what is the output voltage when the two inputs have no differential component. In part, this will be a function of the common mode aspect of the input signals. However, it will be only loosely defined, as under this condition the non-perfect matching across the separate implementations of each FET will have a significant affect. This is not generally of concern as high gain differential amplifiers are used with feedback or as open loop comparators.

6.5.2. NFET Input Operational Amplifier

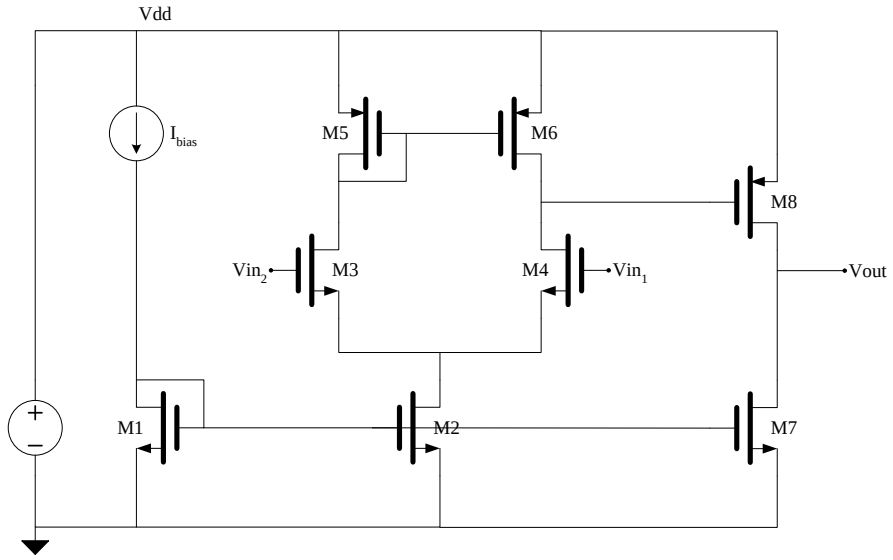


Figure 27: NFET Input, Two-stage, Operational Amplifier

If more gain is needed from that available from the current mirror loaded differential amplifier alone then an additional stage may be added to the output. The signal is single-ended at this stage and so the voltage gain topology of a common source TCA loaded by a current source is used. This circuit is commonly called a two stage operational amplifier.

It should be noted that any high-gain differential amplifier with a high impedance in put and a low impedance output may be considered to be an operational amplifier (differential voltage input and single-ended voltage output). In fact the low output impedance need only come about by feedback for this to be the case (a common analogue design technique). A single stage operational amplifier may be realised by cascoding the load and transconductance devices in a balanced amplifier, thus offering sufficient gain. We shall see later that the important characteristics of an operational amplifier are that it has a dominant low frequency pole and is stable under negative feedback.

The structure of the second stage is chosen as a matter of convenience. The bias reference already exists in the left-hand side of the current mirror (**M1**-**M2**). In terms of DC operation it is normally convenient to dimension **M7** with half the width of **M1** (and **M2**) to lower the systematic input offset voltage. We will see later that this bias is critical to maintaining stability under feedback and will be adjusted. A disadvantage of using a PMOS device in the second stage is that it will have lower transconductance than an NMOS of the same dimensions. Also the NMOS current source load will have a lower output resistance so the voltage gain of the second stage is a compromise for maximum output voltage swing.

In this two stage amplifier, $\boldsymbol{V_{in1}}$ is now the non-inverting input and $\boldsymbol{V_{in2}}$ is the inverting input.

When used in an open loop application, stability is not of concern. For general use, the designer assumes maximum negative feedback is applied when designing the op-amp i.e. an s/c from output to negative input. The issues of stability are dealt with in detail in the Section 7. A summary of the problem is given below.

Two high impedance nodes may be identified in the signal path of this amplifier with negative feedback: one at the gate of **M8** and one at the gate of **M3**. The node at the gates of **M5** and **M6** is also of high impedance but is much less significant. For each of the primary nodes a pole exists in the frequency response of the amplifier. The associated phase shift of 90º per pole means that the sense of the feedback will eventually be inverted as frequency increases. If this occurs when the loop gain is greater than unity, instability will result.

An additional stage or a redesigned second stage may be employed to offer lower output impedance, wider output voltage swing or better output current drive capability. Any additional high impedance nodes will add a pole to the frequency response of the amplifier that must be accounted for by the designer.

6.5.3. PFET Input Differential Amplifier

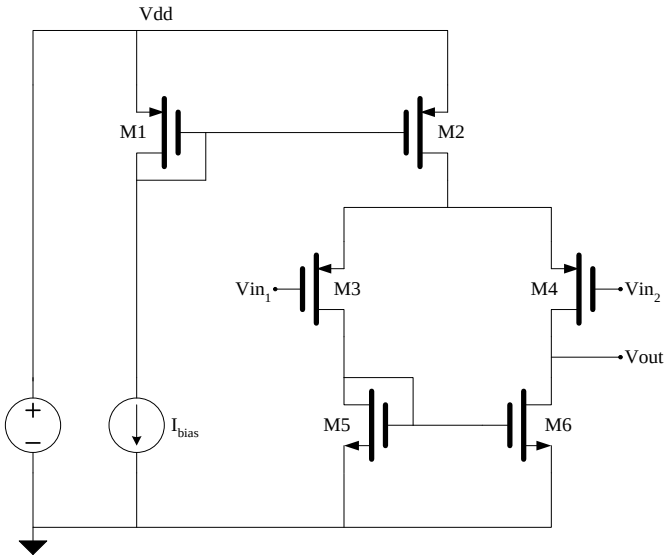


Figure 28: Current Mirror Loaded, PFET Input, Differential Amplifier

The PFET input differential amplifier operates in exactly the same manner as the NFET input structure. However, the bias is a source rather than a sink and a load is formed to the negative supply (earth in this case) using NFET devices. This circuit will operate correctly with input signals ranging to the negative rail but will not function if the common mode aspect of the input signal forces the node at the drain of **M2** high enough to force **M2** out of saturated operation. The converse is true of the NFET input circuit. If all transistor dimensions are the same as the NFET realisation then this circuit will have less gain because the charge carrier mobility of PFET devices is lower (leading to lower transconductance) and the load devices are of lower impedance (leading to poorer conversion of current to voltage).

This circuit has a major advantage over its dual. In a standard N-well process, the PFET devices can share a single well and have their bulk gates connected. This ‘floating well’ results in very closely matched devices which will lower any input offset voltage. Lower gain will result because of the back-gate effect, this is shown below not to be a problem in operational amplifier design. In addition, PFET devices have lower (1/f) noise, so are better placed at the input of an amplifier chain.

If $v_B = v_D$ (The maximum it can be for correct FET operation in saturation) then the device output impedance will be at its lowest. This is partially offset by an increased transconductance.

6.5.4. PFET Input Operational Amplifier

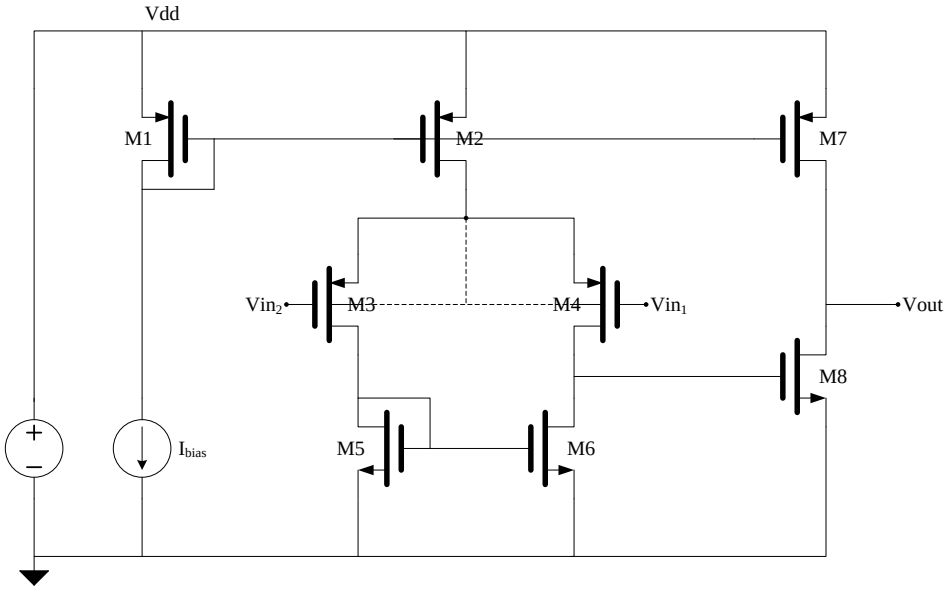


Figure 29: PFET Input, Two-stage, Operational Amplifier

Here the floating N-well back gate connection is shown on **M3** and **M4**. The addition of the second stage follows the discussion above for the NFET input dual of this circuit.

The choice of whether an operational amplifier should be PFET-input or NFET-input is that of the designer. Common considerations in this are outlined below that should lead the reader to note that PFET input operational amplifiers are more common:

- The DC gain is largely unaffected as both have two stages; one of each type of transconductance device.
- PFET input stage op amps offer greater slew rate See [1], Page 228.
- PFET input stage op amps offer greater high frequency gain: see Section 7.2.
- The noise performance of the op amp is dominated by that of the input stage. PFET devices typically offer less 1/f noise but the lower transconductance leads to higher thermal noise. The former is normally the dominant consideration.
- PFET input stage op amps may take the advantage of better component matching by placing the input pair in a floating n-well.

6.6. IC Voltage Reference Circuit

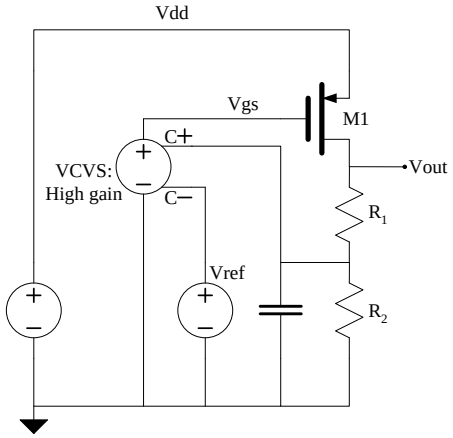


Figure 30: Controllable Voltage Reference Circuit

The IC voltage reference circuit comprises a master voltage reference, a common source PMOS amplifier, an output sensing circuit and feedback via a high gain voltage controlled voltage source (VCVS). Other circuit implementations will perform similarly but the basic topology remains the same. The output voltage is controlled by the values of V_{ref}, R_1 and R_2 and the feedback linearises the response of the circuit to be a function of these parameters. The capacitor is required in order to stabilise the circuit, i.e. to stop the circuit meeting the Barkhousen criterion and may be ignored in DC analysis. The circuit will respond to variations in the current drawn from V_{out} in a manner to keep the voltage fixed. Thus, a low impedance node is formed by circuit action. This need not be the case at V_{ref} because no current need be drawn from it.

On first inspection, it appears that positive feedback is employed as the return path is to the positive terminal of the VCVS. However, this is not the case as **M1** performs an inverting function in signal terms: V_{out} falls as V_{gs} rises.

In order to investigate the operation of this circuit in DC terms, consider the operation of the VCVS in the presence of feedback. This particular VCVS is a high gain device that sets up V_{gs} according to

$$V_{gs} = K(V_{c+} - V_{c-}),$$

where K is a very large positive scalar.

In order for V_{gs} to remain within the bound of V_{dd} and the earth reference, V_{C+}-V_{C-} must be very small. One may say as an approximation that the circuit operation is to force V_{gs} to be whatever is required in order for V_{C+} to equal V_{C-}. This is valid if the gain of the VCVS remains high. (In fact, the difference between the two control inputs is related to the required DC value of V_{gs}.) If the feedback was taken directly from V_{out} then it would track V_{ref}, however, in this case the voltage divider network of R_1 and R_2 add a control parameter to the circuit the operation which may now be approximated by the following linear equation.

$$Vout = Vref\left(1 + \frac{R_1}{R_2}\right).$$

This circuit demonstrates the elegance of feedback in analogue IC design. If the VCVS is implemented by a differential amplifier or an operational amplifier, a circuit comprising a number of active and linear devices would result. Feedback and high gain is exploited to realise a linear function with non-linear components. The stability implication is still to be addressed by the designer.

7. Two Stage Operational Amplifier Design

This circuit is one of the generic building blocks of an analogue architecture realised as an integrated circuit. Here a complete design process is discussed, including the topology and component sizing issues. For this, some background explanation of feedback, poles & zeroes, Bode plots and stability is included.

7.1. Simulation of Feedback Circuits

You are expected to have an understanding of feedback, system stability, poles/zeroes and control systems from your undergraduate study. A good text on the topic, of which there are many copies in the library, is R. C. Dorf, R. H. Bishop, 'Modern Control Systems', 9th Ed. ISBN:0-13-030660-6. A second recommended text is N. S. Nise, 'Control Systems Engineering', ISBN 0-471-44577-0. A limited address is given here which offers the minimum needed for our purpose in this unit.

7.1.1. Simple Feedback System

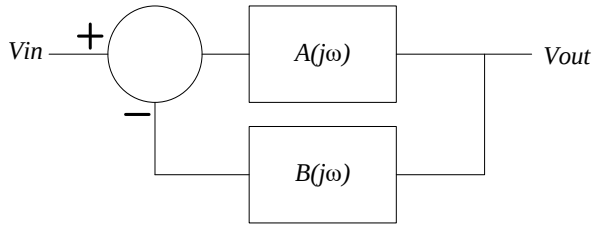


Figure 31: Feedback System

The basic feedback system above comprises frequency dependent gain blocks in both forward and reverse paths and an input subtractor to apply negative feedback.

The open loop gain of this system is given by

$$\frac{Vout}{Vin}(OL) = A_{OL} = A(j\omega)\,.$$

The closed loop gain is calculated thus.

$$\begin{aligned} Vout = A_{CL} &= A(j\omega)(Vin - B(j\omega)Vout) \\ &= A(j\omega)Vin - A(j\omega)B(j\omega)Vout \\ Vout + A(j\omega)B(j\omega)Vout &= A(j\omega)Vin \\ Vout(1 + A(j\omega)B(j\omega)) &= A(j\omega)Vin \\ \frac{Vout}{Vin}(CL) = A_{CL} &= \frac{A(j\omega)}{1 + A(j\omega)B(j\omega)} \end{aligned}$$

This important result affords the designer key metrics to satisfy for stability. A stable feedback system is characterised by the closed loop gain equation providing a finite output over all physical frequencies. Concerning ourselves only with the affect on physical frequencies, the right hand side of the closed loop complex gain equation should be finite, which is to say that $A(j\omega)B(j\omega) \neq -1$.

In practical terms (at least for analogue IC design), the loop gain is examined. This is the gain around the feedback loop which gives the $-A(j\omega)B(j\omega)$ product. When the magnitude of this gain reduces to unity (0dB), the phase of this gain should have not

yet reached zero (migrating from 180º). The amount by which it is greater than zero is called the phase margin. By the same justification, when the phase of this gain reduces from 180º to 0º, the magnitude should be less than unity. The amount by which it is less than unity is called the gain margin. (Note that a subtly different definition exists if the transfer characteristic of the closed loop system is examined rather than the loop gain.)

Alongside the order of the system, the phase and gain margins characterise the time domain response characteristics of the system; for example, the damping factor, overshoot and rise time of the response to a step input. As a guide, it is common for the designer to aim for at least a 70º phase margin in the initial design iteration.

This material should not be new to the reader: [3]-Appendix E gives a good summary of the application to electronic circuits.

7.1.2. Analogue Feedback System with Poles

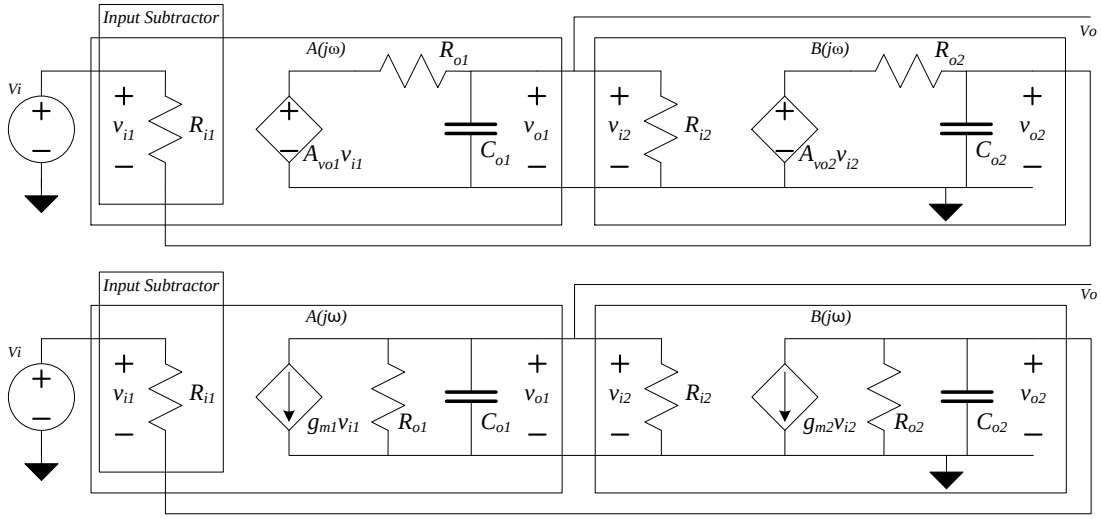


Figure 32: Small Signal Models of an Electronic Feedback System

Each of the gain blocks in the small signal models above has a single pole associated with it. This is a good starting point for analysis as it follows the nature of many amplifier stages in analogue IC design. Commonly many poles and zeroes are present in an amplifier circuit and usually there is one pole that dominates the design process. This is known as the dominant pole and is the lowest frequency pole. At a high-impedance node in a circuit a pole will exist. ***The Management of the dominant poles in a system and, therefore, the management of the high impedance nodes, is key to analogue IC design.***

The ideal controlled voltage source in each amplifier is a s/c in small-signal terms, however, the presence of the resistors $\boldsymbol{R_{o1}}$ and $\boldsymbol{R_{o2}}$ form a high impedance looking back from $\boldsymbol{C_{o1}}$ and $\boldsymbol{C_{o2}}$ respectively. Looking forward from these capacitances, the input resistance of the following stage is seen. If an amplifier is realised by a current source-loaded common source circuit then the output resistor ($\boldsymbol{R_{oX}}$) models the parallel combination of the source-drain resistances of the two devices. From this starting point, the reader may extrapolate the contributing effective resistances of other amplifier structures. The gate resistances may be assumed to be high enough to be ignored in this analysis. The capacitor ($\boldsymbol{C_{oX}}$) models total effective capacitance to ground on that node. This is made up of the physical signal path and the combined capacitances of the FET connections to that line.

Appendix C in [2] offers a good development of the relationship between phase margin and time domain response for such a system.

The key characteristic of a high impedance node to note is that its voltage will vary due to the current being sourced to it. Because there is a capacitance associated with the node, this will charge and discharge which slows the rate of change of this voltage. This physical phenomenon causes the pole in the frequency domain. That is, as frequency increases, the voltage variation at the node will lower. In other terms, a voltage divider circuit is formed by the resistance and capacitance, this will have a frequency-dependent nature.

7.2. Operational Amplifier Stabilisation (Compensation)

Here the specific case of feedback stability applied to the two-stage operational amplifier is discussed.

7.2.1. Compensated Two-Stage Operational Amplifier

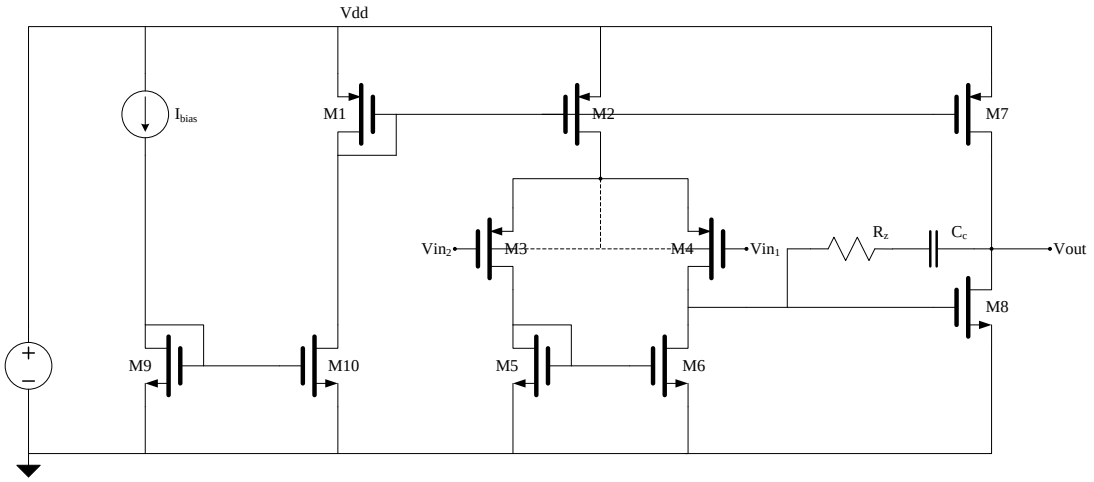


Figure 33: Frequency Domain-Compensated Operational Amplifier

In the circuit above a feedback path has been added from the output of the second stage to the output of the first stage. The capacitor $\boldsymbol{C_c}$ separates the poles associated with each stage in the frequency domain so that one becomes dominant. This is known as 'pole splitting'. Positioning the capacitor across the inverting voltage amplifier formed by the second stage drastically increases its effect on circuit performance relative to a capacitance to ground at this node due to the Miller effect (see Section 6.4.4 in [3] for an explanation). Unfortunately, this also forms a second forward path from the output of the first stage to the output of the circuit. This leads to a positive zero being formed, which has a detrimental effect on the frequency domain characteristic. This is overcome by the presence of $\boldsymbol{R_z}$, which is used to position the positive zero appropriately. The application is explained in detail below.

7.2.2. Small Signal Equivalent Circuit of Two Stage Operational Amplifier

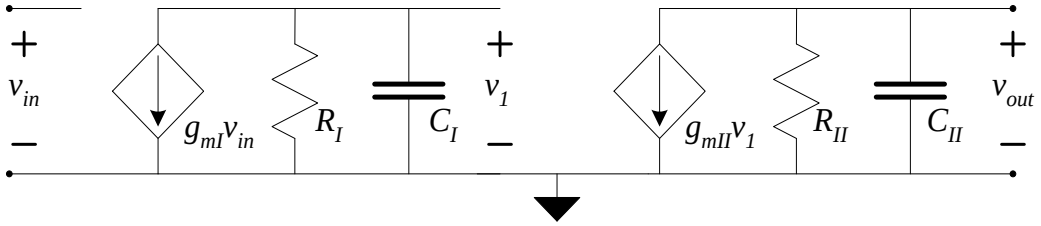


Figure 34: Small Signal Equivalent Circuit

The usual procedure for forming a small signal equivalent circuit gives the generic representation above for the operational amplifier in open loop mode. $\boldsymbol{R_I}$ and $\boldsymbol{C_I}$ are the resistance and capacitance associated with the node at the output of the first stage, and $\boldsymbol{R_{II}}$ and $\boldsymbol{C_{II}}$ are the resistance and capacitance associated with the node at the output of the second stage. The two transconductance parameters follow the same nomenclature format. In this figure, the input voltage is single-ended, but it may be balanced without the loss of generality.

The *S*-plane pole locations for each stage are given by

$$p_1' = -\frac{1}{R_I C_I} = -\omega_I \text{ and } p_2' = -\frac{1}{R_{II} C_{II}} = -\omega_{II}$$

according to the overall transfer function

$$T(s) = -\frac{g_{mI} R_I}{1 + sC_I R_I} \times -\frac{g_{mII} R_{II}}{1 + sC_{II} R_{II}} = \frac{g_{mI} g_{mII} R_I R_{II}}{(1 + s/\omega_I)(1 + s/\omega_{II})}.$$

Note that for physical frequencies

$$T(j\omega) = \frac{g_{mI} g_{mII} R_I R_{II}}{(1 + j\omega/\omega_I)(1 + j\omega/\omega_{II})} = \frac{A_0}{(1 + j\omega/\omega_I)(1 + j\omega/\omega_{II})},$$

where A_0 is the DC gain of the circuit.

In a typical case, the two poles are far away from the *S*-plane origin and relatively close together

7.2.3. Two Stage Operational Amplifier under unity negative feedback

In the absence of a circuit application in which to assess the stability of the operational amplifier under feedback, unity negative feedback is commonly assumed (this is equivalent to making $B(j\omega)$=1 in Figure 31, Section 7.1.1). In this case, the stability analysis requires the evaluation of the equation

$$Loop\ Gain(j\omega) = \frac{-A_0}{(1 + j\omega/\omega_I)(1 + j\omega/\omega_{II})},$$

(this is equivalent to $-A(j\omega)$in Section 7.1.1).

A Bode plot may be formed using standard procedure to estimate the phase and gain margins if circuit parameters are known. When employing the '*stb*' Spectre function a

precise frequency response is delivered from which Bode asymptotical approximations may be extracted if desired.

7.2.4. Construction of Bode Plots of Combined LP and HP Circuits

This section contains material that should be known to a student taking this unit and is included as background reading.

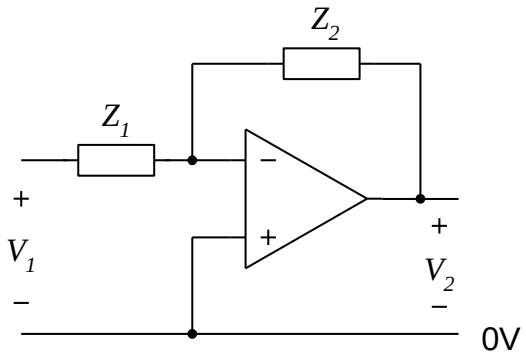


Figure 35: General Form of Inverting Amplifier Based on an Operational Amplifier

Z_1 and Z_2 are impedances and V_1 and V_2 are phasors. Reactive components, *L* and *C*, in Z_1 and Z_2 give the circuit frequency sensitive behaviour. Assuming that the amplifier is ideal, then

$$\frac{V_2}{V_1} = -\frac{Z_2}{Z_1}.$$

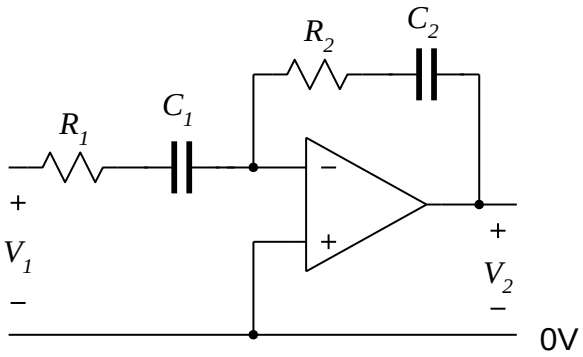


Figure 36: Combined LP and HP Active Filter Using a Discrete Operational Amplifier

Here,

$$Z_1 = R_1 + \frac{1}{j\omega C_1} \text{ and}$$

$$Z_2 = R_2 + \frac{1}{j\omega C_2}.$$

The transfer function may be found and arranged into a standard form for the construction of a Bode Plot, thus:

$$\frac{V_2}{V_1} = -\frac{Z_2}{Z_1} = -\frac{R_2 + 1/\,j\omega C_2}{R_1 + 1/\,j\omega C_1} = -\frac{R_2\, j\omega C_1 + C_1/C_2}{R_1\, j\omega C_1 + 1}$$

$$= -\frac{R_2\, j^2\omega^2 C_1 C_2 + C_1\, j\omega}{R_1\, j^2\omega^2 C_1 C_2 + C_2\, j\omega} = -\frac{C_1(1 + j\omega C_2 R_2)}{C_2(1 + j\omega C_1 R_1)}$$

$$\therefore \qquad \frac{V_2}{V_1} = -\frac{C_1}{C_2}.\frac{1 + j\omega C_2 R_2}{1 + j\omega C_1 R_1} = -K.\frac{N(j\omega)}{D(j\omega)}$$

Of which the amplitude response is given by

$$\left|\frac{V_2}{V_1}\right| = K.\frac{|N(j\omega)|}{|D(j\omega)|}$$

and phase response is given by

$$\arg\left\{\frac{V_2}{V_1}\right\} = \pi + \arg\{N(j\omega)\} - \arg\{D(j\omega)\},$$

where, in both cases, $K = \dfrac{C_1}{C_2}$, $N(j\omega) = 1 + j\omega C_2 R_2$ and $D(j\omega) = 1 + j\omega C_1 R_1$.

The Bode procedure finds piece-wise-linear approximations to the transfer function amplitude and phase characteristics that act as asymptotes as frequency diverges from that defined by the time constants of the system. The accuracy of the approximations reduces as the critical frequencies (e.g. corner or resonant frequencies) are approached, exact solving at these critical points allows the designer to assume a more realistic curve relative to the piece-wise-linear approximation.

Amplitude Response

If we take Log_{10} of both sides of the amplitude equation, then the processes of multiplication and division are replaced by addition and subtraction, respectively. Also, multiplying throughout by 20 gives the result in dB. This process leads to the 'Bode plot' of the amplitude vs. frequency response of a circuit or system if the frequency axis is log-scaled. Thus, the expression becomes:

$$20\log_{10}\left|\frac{V_o}{V_i}\right| = 20\log_{10} K + 20\log_{10}|N(j\omega)| - 20\log_{10}|D(j\omega)|\,.$$

Note that whilst the independent variable is termed 'frequency' it is expressed as angular velocity.

Consider firstly the numerator term. It is convenient to replace $C_2 R_2 = 1/\omega_2$.

Thus

$$20\log_{10}|N(j\omega)| = 20\log_{10}|(1 + j\omega/\omega_2)| = 20\log_{10}\sqrt{1 + (\omega/\omega_2)^2} \text{ dB.}$$

Consider the extremes of independent variable with respect to the constant ω_2.

(i) when $\omega << \omega_2$, $20\log_{10}|N(j\omega)| \approx 0\,\text{dB}$

(ii) when $\omega >> \omega_2$, $20\log_{10}|N(j\omega)| \approx 20\log_{10}(\omega/\omega_2)\,\text{dB}$

Thus, $20\log_{10}|N(j\omega)| = 20\log_{10}\omega - 20\log_{10}\omega_2$ dB

Plotting $20\log_{10}|N(j\omega)|$ vs. $\log_{10}\omega$ gives a straight line, with a slope of 20 (dB/decade), passing through the point $(\omega_2, 0\text{dB})$. In practice, we use graph paper having a logarithmic scale for ω. Thus

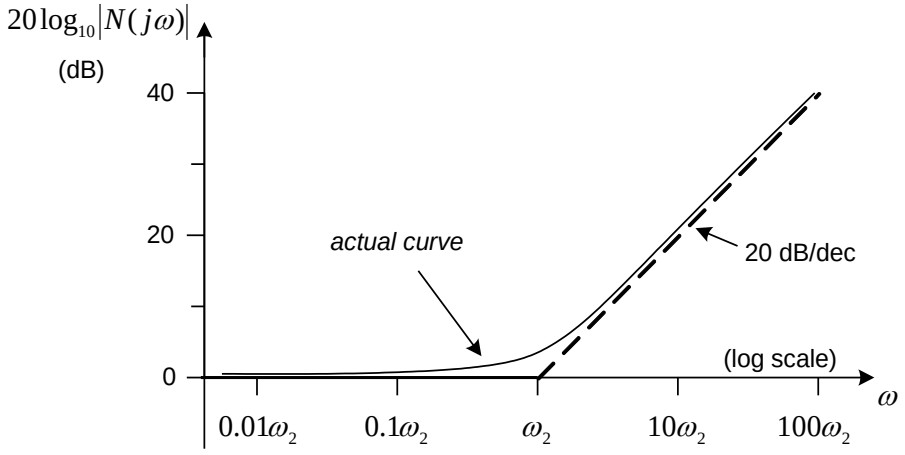


Figure 37: Bode Plot of $20\log_{10}|N(j\omega)|$

We have defined the low-frequency and high-frequency asymptotes respectively of $20\log_{10}|N(j\omega)|$. These meet at $\omega = \omega_2$, where the actual value of the function is:

$$20\log_{10}|N(j\omega)| = 3\text{dB}$$

We can repeat this process and plot $-20\log_{10}|D(j\omega)|$ vs. $\log_{10}\omega$. Thus,

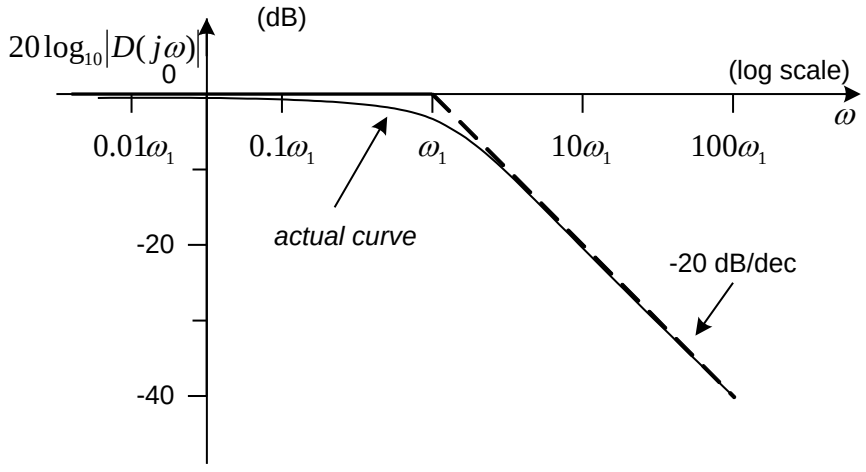


Figure 38: Bode Plot of $-20\log_{10}|D(j\omega)|$

The reason for plotting $-20\log_{10}|D(j\omega)|$ is that we can build up the Bode Plot of the amplitude *vs*. frequency response by simply adding together the component parts.

Suppose, in this example, $C_1 = 10C_2$, then $20\log_{10}K = 20\text{dB}$. Also, suppose that $\omega_2 > \omega_1$ (i.e. $C_2R_2 < C_1R_1$). The complete response given below shows a low

frequency gain of 20dB with a reducing gain for $\omega_2 > \omega > \omega_1$ and a constant attenuation when $\omega > \omega_2$.

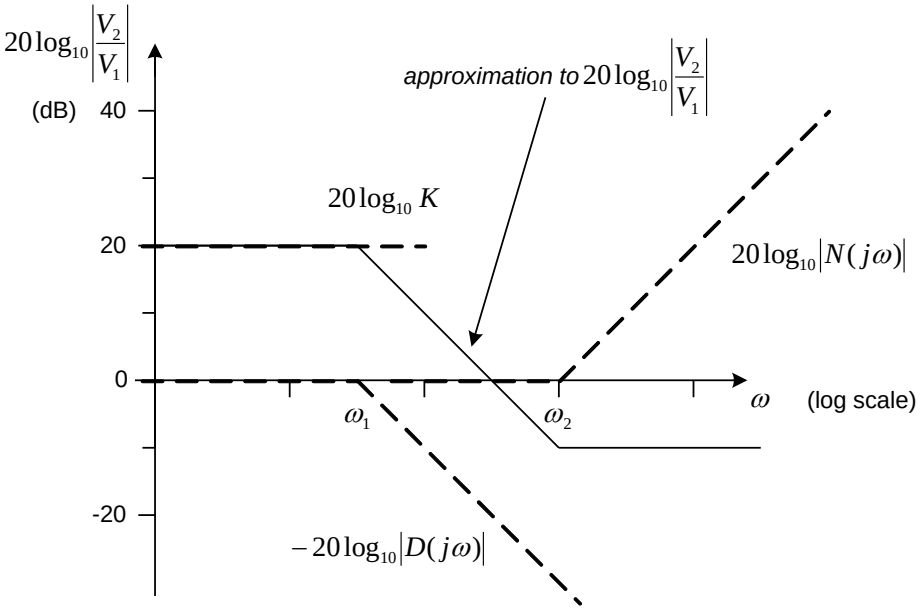


Figure 39: Example Amplitude Bode Plot for Low Pass Filter Function

If now, we design $\omega_2 < \omega_1$, the circuit offers a high pass function. As shown below.

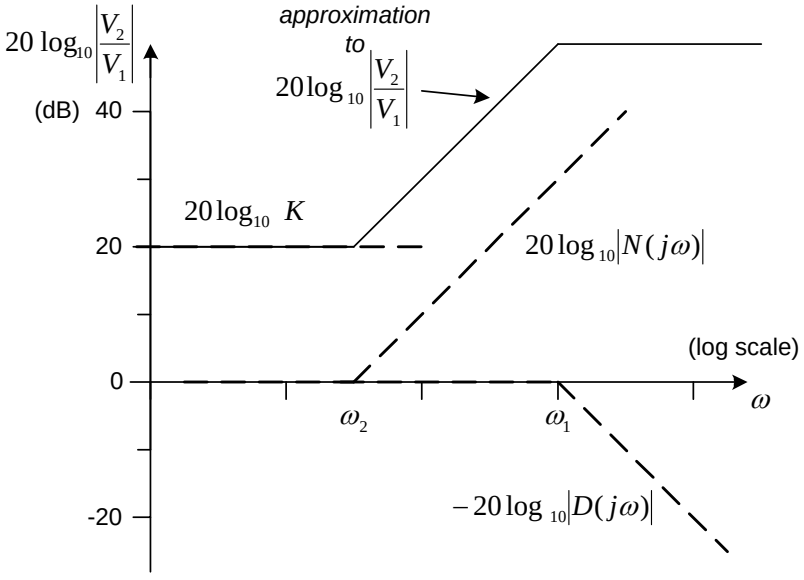


Figure 40: Example Amplitude Bode Plot for High Pass Filter Function

Phase Response

We can build up the circuit phase response in the same way from the component parts of the phase response equation. Consider the numerator term:

$$\vartheta_N = \arg\{N(j\omega)\} = \arg\{1 + j\omega/\omega_2\} = \tan^{-1}(\omega/\omega_2).$$

It may be seen that when $\omega << \omega_2$, $\vartheta_N \to 0$,

when $\omega = \omega_2$, $\vartheta_N = \pi/4$,

and when $\omega >> \omega_2$, $\vartheta_N \to \pi/2$.

The graph of ϑ_N against ω, using a log frequency scale, can be approximated by three straight line segments, as shown below.

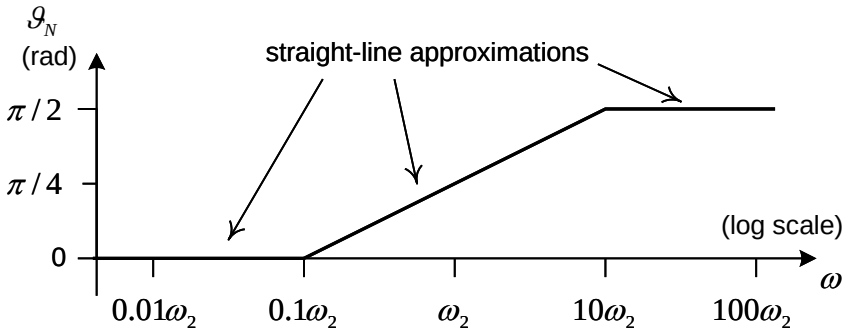


Figure 41: ϑ_N vs ω

Similarly, putting $\vartheta_D = \arg\{D(j\omega)\}$, the form of $-\vartheta_D$ vs ω is shown below,

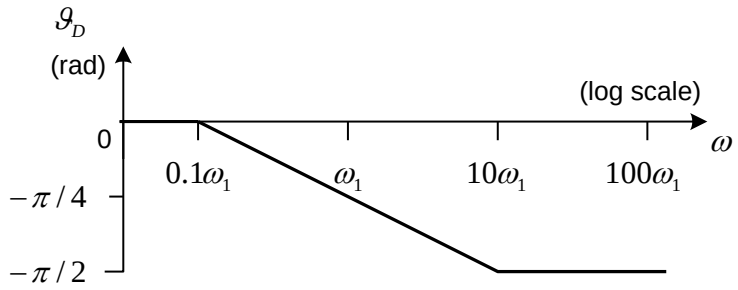


Figure 42: $-\vartheta_D$ vs ω

The resultant plot of $\vartheta_N - \vartheta_D$ vs ω is shown below.

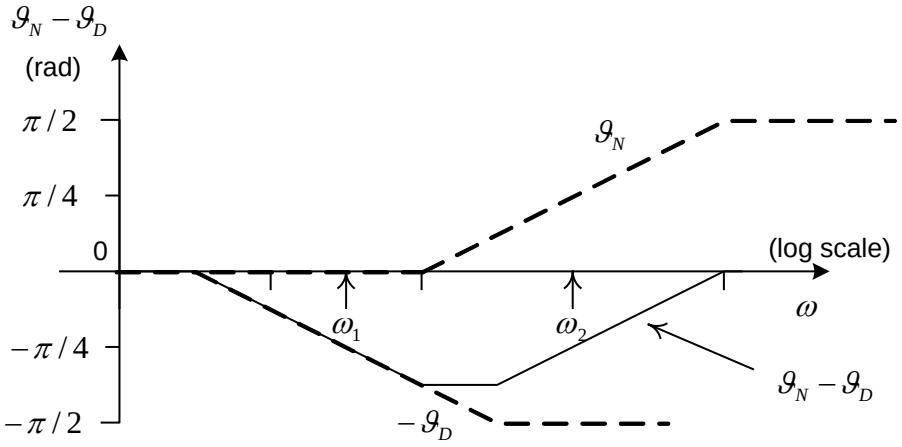


Figure 43: $\vartheta_N - \vartheta_D$ vs ω

The complete phase response for the treble cut circuit is obtained by adding π to the response shown above.

7.2.5. Pole Splitting by Miller Compensation

The addition of the compensating capacitor, C_c, is modelled in the small signal equivalent circuit as below.

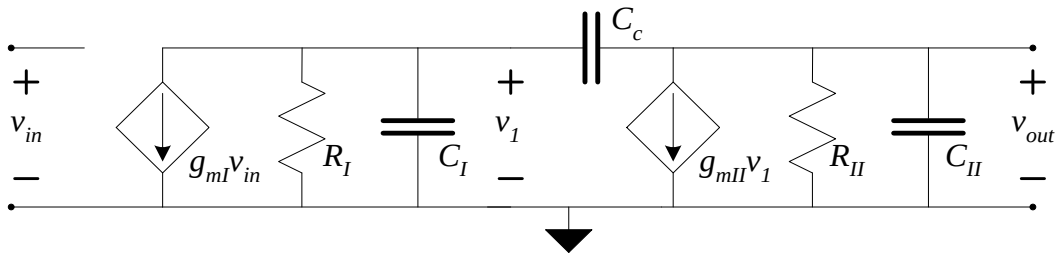


Figure 44: Small Signal Equivalent Circuit with Compensating Capacitor

This has the following affect on the pole locations:

- The effective capacitance in parallel with C_I is increased by an additive of approximately $g_{mII}R_{II}C_c$ due to the Miller effect. This moves p_1' to p_1, a point much closer to the origin of the *S*-plane.
- The negative feedback applied to the second stage moves p_2' to p_2, a point further away from the origin of the *S*-plane. This is because the impedance at this output node is lowered.

Page 257 of [2] rigorously demonstrates how the new pole locations are calculated.

The complete expression for the transfer function of the two stage operational amplifier with compensating capacitor is

$$\frac{V_o(s)}{V_i(s)} = \frac{g_{mI}g_{mII}R_IR_{II}(1-sC_c/g_{mII})}{1+s[R_I(C_I+C_c)+R_{II}(C_{II}+C_c)+g_{mII}R_IR_{II}C_c]+s^2R_IR_{II}[C_IC_{II}+C_cC_I+C_cC_{II}]}$$

For our purpose here it is sufficient to note that the new pole locations are very closely approximated by

$$p_1 \cong -\frac{1}{g_{mII}R_IR_{II}C_c} \text{ and } p_2 \cong -\frac{g_{mII}}{C_{II}}.$$

Comparing the location of p_1 with $p_1' = -\frac{1}{R_IC_I}$, it may be seen that the location remains on the real axis of the *S*-plane but has a much reduced magnitude.

Comparing the location of p_2 with $p_2' = -\frac{1}{R_{II}C_{II}}$, it may be seen that the location remains on the real axis of the S-plane but has a much increased magnitude.

Pole splitting has been achieved. In order to understand the implication to the stability of the device under unity negative feedback, consider the loop gain equation which will merely be

$$Loop\ Gain = -\frac{v_{out}}{v_{in}} \text{ (OL)}.$$

This expression will have large amplitude and a 180º phase shift at low frequency that migrates towards small amplitude and a smaller phase shift as frequency increases.

Because of the distance between the poles, p_1 is said to be dominant and defines the operation of the device over the greater part of the frequency space. The low frequency dominant pole will cause a roll-off of magnitude response of 20dB/decade from (near) zero Hz. The total phase shift associated with this pole is 90º. The aim is to force the gain response of the loop gain to cross 0dB far enough below the frequency of the second (non-dominant) pole such that the phase shift associated with it does not pull the overall phase shift unduly close to 0º. The closer to 0º phase at this crossing point, then the more tendency the closed loop circuit will have towards transient ringing and overshoot. Instability results if the loop gain response transcends 0º whilst the gain response is above 0dB.

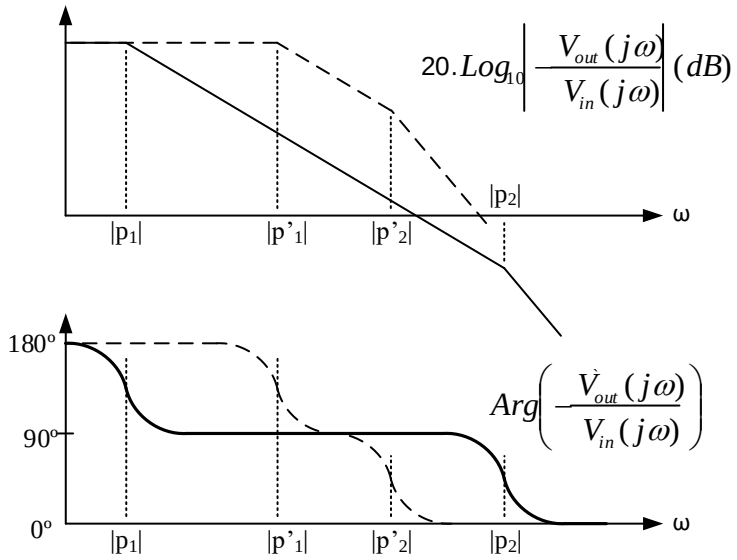


Figure 45: Pole Splitting in the frequency domain

The frequency at which the gain of the operational amplifier reduces to 0dB is called the unity gain bandwidth (GB). The GB is assumed to also be the relative bandwidth of the device as it has DC gain and the dominant pole is near DC. Note that we are using logarithmic-scaled angular frequency axes.

By analysing the equation for $-\frac{V_o(s)}{V(s)}$ and substituting $S = j\omega$ for physical frequencies it is shown that the angular velocity at which the gain is equal to unity is

$$\omega_{0dB} = GB = \frac{g_{mI}}{C_c} \quad rs^{-1}.$$

Unfortunately, this pole splitting technique introduces a right-half plane (RHP) zero at

$$z_1 = \frac{g_{mII}}{C_c}.$$

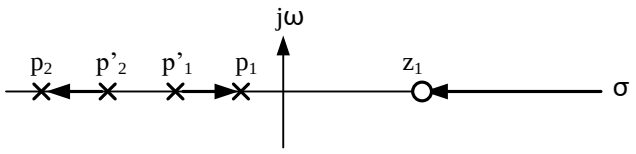


Figure 46: Pole Splitting with RHP Zero on the S-plane

Such zeroes are most unfriendly to the analogue designer as they have both the stability disadvantages of a LHP pole (pulling the overall closed loop phase shift away from 180º and towards 0º) and a RHP zero (increasing the magnitude). The zero is formed by the two paths from the input of the second stage to the output of the circuit. One path is via the output transconductance stage and the second is via $\boldsymbol{C_c}$. At some complex frequency the magnitude of these two paths will be equal and their phase opposite; forming the zero.

For a designed phase margin of 60º, for an operational amplifier with a large $A_v(0)$, it may be shown that,

$24.3^{o} \approx \tan^{-1}\left(\frac{GB}{|p_2|}\right)$ which gives

$$|p_2| \geq 2.2GB$$

$$\frac{g_{mII}}{C_{II}} \geq 2.2\frac{g_{mI}}{C_c} \qquad \text{(a)}$$

60º is a compromise between overshoot and response time. In reality, this should be greater (say 75º) to account for PVT variation. The maximum possible phase margin is 90º due to the effect of the low frequency dominant pole. Commonly in CMOS designs, a phase margin greater than (say) 85º will require a low frequency gain that is insufficient or a compensating capacitor so large that the RHP zero is a prolem.

In order to have negligible effect on the phase at the GB, the frequency of the zero should be at least ten times higher, leading to

$$|z_1| \geq 10.GB$$

$$\frac{g_{mII}}{C_c} \geq 10.\frac{g_{mI}}{C_c}$$

$$g_{mII} \geq 10.g_{mI} \qquad \text{(b)}$$

Therefore, combining (a) and (b), $C_c \geq 0.22\, C_{II}$. Note that $\boldsymbol{C_{II}}$ is dominated by $\boldsymbol{C_L}$ (the load capacitance at the output of the circuit).

7.2.6. Control of Right-Half Plane Zero

The most common technique for controlling RHP zero is to add a resistor $\boldsymbol{R_z}$ in series with $\boldsymbol{C_c}$. The reader is again referred to [2] for a formal derivation, however, here it is sufficient to know that the resistor forms the following approximate roots assuming $\boldsymbol{R_z}$ is small compared with $\boldsymbol{R_I}$ and $\boldsymbol{R_{II}}$, and the poles are widely spread in the frequency domain.

$$p_1 \cong -\frac{1}{g_{mII} R_I R_{II} C_c}$$ (as before)

$$p_2 \cong -\frac{g_{mII}}{C_{II}}$$ (as before)

$$p_4 = -\frac{1}{R_z C_I}$$ (new non-dominant pole)

$$z_1 = \frac{1}{C_c(1/g_{mII} - R_z)}$$ (new position for RHP zero)

Note that there will also be a non-dominant pole due to the high impedance node associated with the current mirror load of the first stage (usually referred to as p_3).

$\boldsymbol{R_z}$ does not appear in the approximate equations giving the positions of the two most dominant poles in the *S*-plane. Thus, it may be considered as a predominantly independent controller for the zero position.

The RHP zero may be removed completely by setting $R_z = {}^{1}/_{g_{mII}}$ (i.e. placed back at infinity). Alternatively, it may be moved to the LHP and placed on top of p_2. However, this is not often attempted due to the additional phase problems caused if this position is not exact (consider the difficulty of matching a transconductance to a conductance in a CMOS design). More commonly, it is moved higher in frequency than the gain bandwidth (GB) by at least a factor of ten.

7.2.7. Slew Rate

The slew rate is the maximum rate at which the output can change. Therefore, it is not a wholly frequency-domain metric as the output amplitude is also of concern. An analysis of the large signal circuit given in Figure 33 (Section 7.2.1) shows that the maximum current available to charge or discharge $\boldsymbol{C_c}$ is the tail current of the first stage differential amplifier, $\boldsymbol{I_{M2}}$.

The charge associated with a capacitor is given by

$$q = CV\,.$$

The current through a capacitor is the rage of change of charge and is, therefore, given by

$$I = C\frac{d}{dt}V\,.$$

We may state that $Max\left(\left|\frac{d}{dt}V\right|\right)$ is the slew rate (SR), and so

$$SR = \frac{I_{M2}}{C_c}\ vs^{-1}.$$

7.3. Operational Amplifier Design Summary

In this section a 'cook-book' approach to operational amplifier design against a set of specifications is given. Practice with this method will result in a full understanding of the trade-offs in operational amplifier design. The following two-stage operational amplifier topology is assumed.

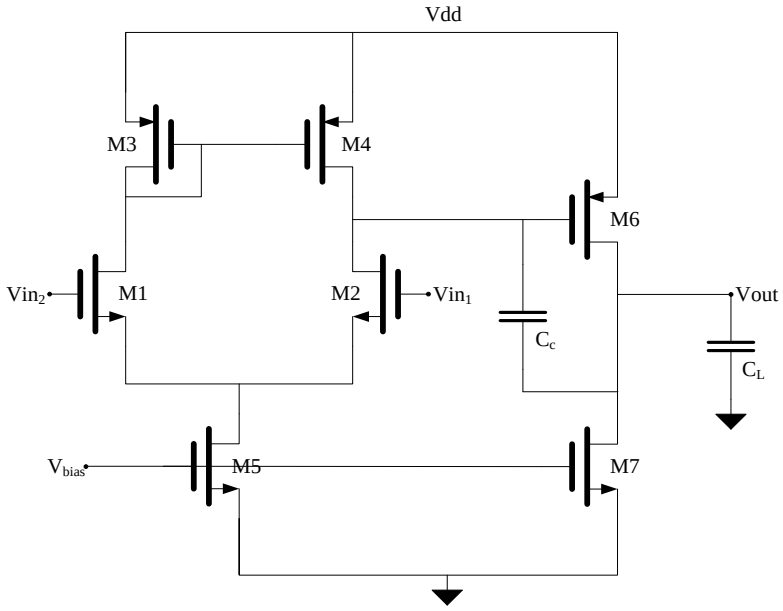


Figure 47: Two-Stage Operational Amplifier for 'Cook-Book' Design

7.3.1. Assumptions, Specifications and Design Equations

Assumptions and Nomenclature

FET gate dimensions: We define $S_i=W_i/L_i$ and $I_i=I_{Mi}$ for transistor **Mi**.

FET transconductance and output conductance:

We extract

$g_{mI}=g_{m1}=g_{m2}$

$g_{mII}=g_{m6}$,

$G_I=1/R_I=g_{ds2}+g_{ds4}$

and $G_{II}=1/R_{II}=g_{ds6}+g_{ds7}$.

Parameter Specifications

The following parameters are commonly given to the designer as a specification.

- DC Gain, $A_v(0)$
- Gain Bandwidth, GB
- Load Capacitance, C_L
- Slew Rate, SR
- Input common mode maximum, ICMR+
- Input common mode minimum, ICMR-

- Output Voltage Swing

Design Equation Summary

Critical design equations are given below (gain parameters are small-signal).

Slew Rate: $SR = \frac{I_5}{C_c}$

First Stage Gain: $A_{v1} = \frac{-g_{mI}}{G_I} = \frac{-g_{m1}}{g_{ds2} + g_{ds4}} = \frac{-2g_{m1}}{I_5(\lambda_2 + \lambda_4)}$ (Note $g_{ds} = I_D.\lambda$)

Second Stage Gain: $A_{v2} = \frac{-g_{mII}}{G_{II}} = \frac{-g_{m6}}{g_{ds6} + g_{ds7}} = \frac{-g_{m6}}{I_6(\lambda_6 + \lambda_7)}$

Gain Bandwidth: $GB = \frac{g_{mI}}{C_c}$

Output pole: $p_2 = \frac{-g_{mII}}{C_L}$

RHP Zero: $z_1 = \frac{g_{mII}}{C_c}$

7.3.2. Design Procedure

1) Choose **L**: Choose a nominal device length for the design based on the required speed. Within this unit we will use a relatively long channel (1μm) to ensure that the hand calculations are accurate. Use the following equation to find f_T (transition frequency):

$$f_T = \frac{1}{2.\pi}.\frac{3}{2n}.\frac{\mu}{l^2}(V_{GS} - V_T)$$

2) Choose C_c: Given C_L, choose C_c given by $C_c \geq 0.22C_L$.

3) Choose I_5: The tail current of the first stage defines the slew rate and is given by $I_5 = C_c.SR$.

4) Choose S_3 (= S_4): If ICMR+ is specified: the width of **M3** should be large enough such that **M3** remains in saturation at this maximum input. If ICMR+ is not specified then a good first pass is to make S_3 = S_4 =1 as a compromise between the input common mode maximum value and capacitance at the mirror gate node. This capacitance should be kept relatively small in order to ensure that p_3 is not of concern in the design. It should be noted that this device scaling will result in a poor ICMR+ result and may need to be compromised.

5) Choose S_1 (= S_2): Dimension the width of the input devices such that the GB requirement is met according to the equation $g_{mI} = C_c.GB$.

6) Choose S_5: Given I_5 and the specified ICMR-, choose W_3: this should be large enough such that **M5** remains in saturation at this minimum input.

7) Choose g_{mII}: For a 60º phase margin choose

$$g_{mII} = 2.2 g_{mI} (C_{II} / C_c)$$

8) Choose S_6 and I_6: For a minimum systematic offset, $V_{GS4}=V_{GS6}$ (preserving the current mirror symmetry) and so

$$S_6 = S_4 \frac{g_{mII}}{g_{mI}} = S_4 \frac{g_{m6}}{g_{m4}} .$$

Knowing g_{m6} and S_6, I_6 may be found, however, there is a degree of freedom between the component geometry and the current. This is adjusted so that the output maximum voltage swing is accommodated. Increasing S_6 and reducing I_6 will increase this value.

9) S_7 may now be found such that I_6 is correct given V_{bias}.

10) The overall gain may now be checked from

$$|A_{v0}| = \frac{2 g_{m1}}{I_5(\lambda_2 + \lambda_4)} \cdot \frac{g_{m6}}{I_6(\lambda_6 + \lambda_7)} .$$

If the gain is insufficient, I_5 and I_6 may be decreased and/or S_2 and/or S_6 increased. In this case the design equations should be rechecked. The table in Section 7.3.3 gives an insight as to how the parameter shifts affect the overall circuit performance.

In all of the steps above, formal design equations exist; however, the reader should use the simulation environment to explore the design procedure. At all stages, if a particular requirement cannot be met then iteration and specification review should be applied. The procedure is appropriate for implementations that are nowadays termed *long channel*, (say, in excess of 250nm). Many of the design equations become much more complicated for post 2008 *short channel* implementations (say, below 90nm).

7.3.3. Performance Dependence on Key Circuit Parameters

It is left as an exercise to the student to prove the content of the table below that summarises the affect of critical circuit parameter choices on the key metrics of performance of the operational amplifier.

	DC current		M1 and M2		M3 and M4		Inverter	Inverter load		Compensation Capacitor
	I_5	I_7	S	L	W	L	S_6	W_7	L_7	C_c
Increase DC gain	√↓	√↓	√↑	↑		↑	√↑		↑	
Increase GB	√↑		√↑							↓
Increase RHP Zero		√↑					√↑			↓
Increase SR	↑									↓
Increase C_l										↑

8. Simulation Objectives and Design for Manufacture

We now have a circuit and a test-bed. This section discusses how it is to be evaluated with respect to what is likely to be received back from the fabrication facility.

8.1.Nominal Performance

Under this condition, the parameters used in the SPICE models of the components are nominal. For example the mean of the process bands for doping concentration, dimensions, metal thickness and carrier mobility are assumed. Further, an ideal voltage rail DC term is assumed alongside a fixed temperature. This is a good starting condition for integrated circuit design as it greatly simplifies the procedure. However, once a circuit topology is established by simulation at nominal performance, more realistic fabrication and operational restraints should be taken into account.

8.2.Process, Voltage and Temperature Corner Analysis

8.2.1. Process Corners

This form of analysis is suitable for both analogue and digital circuits; indeed it is not normal to go beyond this (i.e. to statistical analysis) for designs that contain only digital circuits.

For FET process corners, it is usual to refer to the expected extremes of operation in terms of ‘Fast’ implying greatest Small Signal Transconductance, and ‘Slow’ implying least Small Signal Transconductance. There will be several process parameters and photolithography variations that cause the change in transconductance; these are all set at the appropriate end of their expected process band to set the overall g_m characteristic of the device top be either fast or slow.

Where a process includes resistors (standard digital processes do not), the fast/slow convention is extended to imply conductance extremes.

The situation is further complicated by the fact that NFETs and PFETs should be considered separately. This is because as some of the process and realisation parameters vary to drive a NFET to its fast state will, at the same time drive a PFET to its slow state. Thus, at a NFET fast corner, there are two possible corners for a PFET; fast and slow and visa versa.

It is normal to refer to a particular FET corner in the order NFET-PFET, thus, FS implies fast N and slow P. The four corners for FETs are thus FF, FS, SF & SS.

It is important to note that changing one particular corner state may (and generally does) affect more than just the referred component class. For example, the performance of a NFET will be different for corners FF and FS even though the NFET corner is unchanged. This is because the process parameters and physical extremes for differing devices are not mutually exclusive.

8.2.2. Voltage Corners

Power supply voltages will vary in DC terms (AC signals on the power supply are addressed separately by PSRR analysis). The circuit should be simulated at the expected extremes of absolute value. It is standard (commercial) design practice to allow an input

power rail margin of **[Specified Value]+/-10%**; where there is no internal power conditioning or regulation on the circuit this provides the two voltage corners.

8.2.3. Temperature Corners

Integrated circuits aimed at the commercial market generally assume the nominal junction temperature to be 60ºC (the temperature at the channel in FETs). However, in operation, this may vary drastically, not only due to ambient temperature variation but also due to thermal energy dissipation in the die. It is standard (commercial) design practice to assume -40ºC and +125ºC as the operational temperature corners.

8.2.4. Corner Simulations

Assuming a process contains resistors alongside the two FET forms; there are eight process corners, two voltage corners and two temperature corners. Each combination of corners should be simulated and the functionality of the circuit ensured. Therefore, there are thirty-two corners to be simulated in addition to the nominal values. High yield circuits may also be checked using nominal values as third corners for all PVT values. Note that a process may contain corner models for capacitors and diodes in addition to FETs and resistors. BiCMOS processes will also require analysis for the bipolar corners. It can be seen that the parameter-space becomes huge unless some degree of rationalisation at the sub-circuit level is employed.

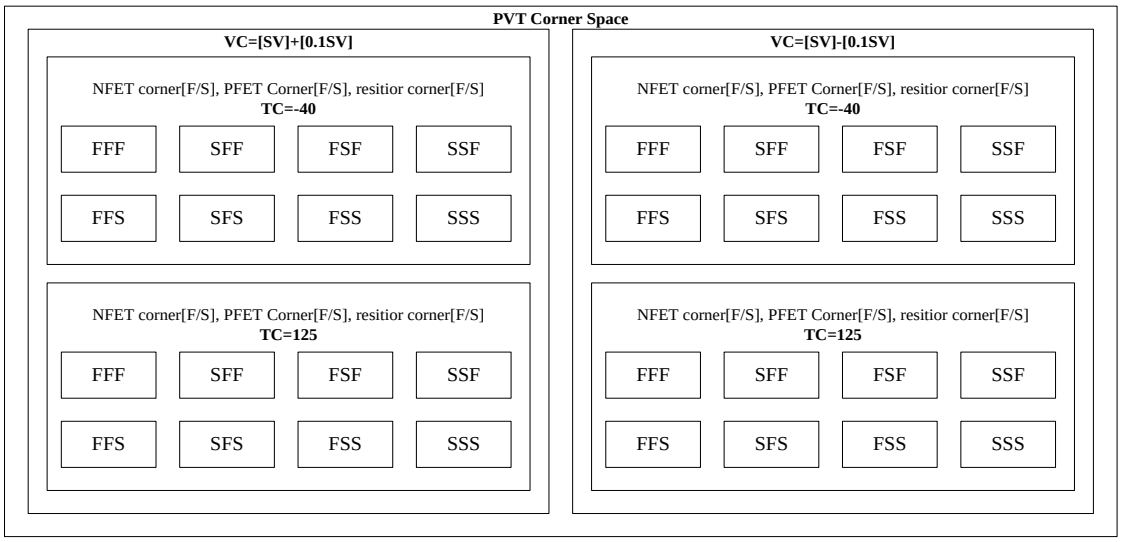


Figure 48: 32 process corners for a CMOS process with resistors.

8.3.Statistical Component Variation Analysis

Corner analysis is limited to the extremes of the key result of process parameter variation, this not sufficient to build up a complete picture of the statistical likelihood of an analogue circuit passing a functionality hurdle. Hence, there is a need for simulation based on an appropriately distributed variation of each process parameter (usually Gaussian). Initially this is carried out at nominal values of voltage and temperature and later extended to process parameter variation at the **VT** corners. This is commonly referred to as Monte Carlo (**MC**) Analysis because of the random mix of conditions for each simulation run. A number of Monte Carlo runs (simulations) are performed and key circuit metrics recorded in order to build up a statistical impression of expected circuit performance.

It is the decision of the designer as to what the acceptable performance metric is and its required attainment probability. However, if a sub circuit has a probability of functionality of 98% and there are 20 such sub circuits in a system then the overall fabrication failure rate is 1/3 (assuming mutually independent failures). There will always be further die failure due to contamination during manufacturing.

Because it is not common practice to carry out **MC** analysis on digital circuits, a manufacturing process aimed solely at the digital market will not offer the **MC** statistical models.

8.4.Designing a Robust Analogue Circuit

Digital circuits may be designed at the minimum geometry for a given process as their operation is based on a switching function between two extremes. Because of this bi-state operation the circuits are inherently robust to process variations (also **VT**). Indeed, digital design is usually carried out via high-level hardware description languages, such as HDL and Verilog that ultimately refer to a standard family of logic cells defined by the process supplier. All of these cells have been previously designed at the schematic level and laid out so as to occupy minimum area on the die.

The dimensional variations caused by the photolithography doping, and etching processes in integrated circuit manufacturing are (approximately) fixed in absolute terms. Therefore, as the geometry of a feature is increased, its dimensional percentage tolerance reduces. It is common practice in analogue IC design to employ dimensions much larger than the minimum geometry so that the circuit performs within requirements across PVT and Statistical Analysis. The shortcoming of increasing the size of the FETs in a design is the lowering speed that results (due to the gate capacitance increase); this is a design trade-off for a given process.

The global market for ICs is dominated by digital solutions. It is a cruel fact that the analogue circuit designer has to use processes that are, in the greater part, merely digital but with process stage additions to provide components such as resistors and capacitors. The fabrication houses offering these processes rarely offer useful statistical data in its SPICE models and in some cases, in-house statistical models are generated based on a best-engineering-guess. Additional problems arise from this trend as with geometry reduction, power supply voltage reduces. In many cases, the classic analogue IC circuit topology may no longer be used (e.g. the telescopic cascode differential amplifier).

Commonly the digital parts of an IC will run from the lowest specified power supply voltage. Processes that support analogue design may also allow one (or sometimes two) higher voltage supplies on the same die. Higher voltage FET devices are included in the design kit that will have minimum geometries significantly larger than that defined by the process minimum. With these, some of the problems of designing an analogue circuit on a digital process are alleviated.

Beyond increasing the dimensions of the components on the die, matching plays a key role in robust analogue circuit design. This topic is addressed in Section 10.2.

9. SPICE Simulation Using Spectre

We now have a circuit and a test-bed and an understanding of the practical remit of simulation. This section discusses the ways in which we investigate the performance of the circuit.

The standard method of investigating the operation of circuits is to resolve node voltages and branch currents. In essence, establishing the force magnitudes (voltage) and flow magnitudes (current) across a circuit will tell the designer a great deal of what he/she needs to know. However, manually putting this information together to form an impression of the circuit response to varying bias and input conditions is not necessary or practicable. SPICE (Simulation Package with Integrated Circuit Emphasis) is the tool used to do this matrix resolution. Some SPICE tools are open source but many proprietary versions, with associated claims of improvement, exist. Cadence's version of SPICE is Spectre with an extension of functionality for use at high frequency called SpectreRF. Within Cadence, the platform on which simulation using Spectre (and other analogue circuit resolution algorithms) is carried out is called the Analogue Design Environment (ADE).

9.1. DC Sweep for Baseline Circuit Performance

A single static investigation of the node voltages and branch currents will result in an operating point result. This is a useful tool to make sure the operating mode of the active components in a design is correct and that power, bias and common mode voltages are correct. However, much greater information is offered to the designer if a sweep is carried out and the outputs (either node voltages or branch currents) is plotted against the independent swept variable. The independent swept variables may be:

- Temperature: The global variable that defines the operating temperature of all instances of components in a design is swept.
- Design Variable: A variable defined in the schematic (and also in ADE) is swept. This variable may be used in different ways via mathematical expressions in a number of components. This may be a preferable method than sweeping a component parameter directly (below) as an intuitive label will appear on the output graph independent scale.
- Component Parameter: A parameter in a particular instance of a component is swept.
- Model Parameter: A parameter in all instances of a model (component) are swept.

The number of circuit operation calculations per sweep may be configured by the designer. It is always best not to rely on the default settings as only full control will give the designer confidence in the result.

9.2. Transient (time domain)

With a transient analysis the operation of the circuit is calculated at sequential time instances with the accuracy (number of observations per unit time) under the control of the designer. For this simulation, a time-varying stimulus is appropriate.

Varying the bias conditions of a circuit in the time domain and simulating with a transient analysis will give a comparable result to that of a DC sweep as long as the sweep is slow enough for the circuit to stabilise at each observation time. However, the generality of the analysis is lost and so it will not provide a valid result.

9.3. AC Sweep (frequency domain)

Frequency domain analysis using an AC sweep is a small signal evaluation only. For this analysis to be valid the Bias (DC) operating point of the circuit must be fixed. This simulation works by linearising the performance of the circuit at an operating point and assuming a small signal perturbation from a stimulus that is swept in frequency as defined by the designer. As the circuit is linearised, it does not matter what the magnitude of the AC stimulus is but unity is favoured as this will give an output graph that is the transfer function from stimulus to selected output.

Note that certain circuit stimuli such as *vdc*, *vsin* and *vpulse* have AC parameter fields; it is these fields that are used by the AC simulation algorithm. Importantly, in the *vsin* component (and other similar ones) this is not the same as the magnitude of the sinusoid that is generated in the time domain.

9.4. Parametric Analysis for Family-trace Results

Simulations may be nested. That is, a number of swept variable analyses may be performed at a given set of parameter conditions. By this method, a family of output traces may be observed on the same graph against the same independent variable for a range of (for example DC bias) conditions.

9.5. Stability Analysis in Feedback-controlled Circuits

The stability analysis tool facilitates the investigation of the frequency domain characteristics of any feed back loops in a circuit. This cannot be done with the AC analysis tool as there should not be an input stimulus. The Characteristics of a feedback loop may be analysed over a number of swept variables but frequency is the most common with parametric analysis used to account for variable DC (bias) conditions and PVT.

A probe in the form of a Null-configured *vdc* source is inserted into the feedback loop and the loop gain transfer function is established over the sweep range. The output is best observed via a Bode plot; phase and gain stability margins may be observed. The stability margin is at the discretion of the designer but it is common to aim for at lest 75 degrees (60º minimum).

9.6. Application of Monte Carlo and Corner Analyses

9.6.1. Corner Analysis

Process corners are managed by addressing separate sections in the Process Design Kit Library. This will always include NFET and PFET devices (Fast/Slow = ‘all design factors that lead to higher transconductance’ / ‘all design factors that lead to lower transconductance’), and may also include Polysilicon resistors (Fast/Slow = high conductivity/low conductivity). For example a library section sf_f will give models of

FETs and Resistors that are slow(NFET), Fast(PFET) and fast(PolyRes). Follow the menu structure **Setup|Model Libraries** in the ADE to set a process corner.

Voltage and Temperature corners are managed directly in the ADE, following the menu structure **Setup|Temperature** facilitates the setting of temperature at the simulator level. The global variable 'temp' may also be used in the parametric analysis, or alternatively, accessed directly in a DC analysis as a swept variable. Voltage corners are accessed by setting an appropriate design (or component) variable in the ADE such that the circuit is driven by alternate DC voltage source values.

9.6.2. Monte Carlo Analysis

Monte Carlo analysis (also known as Analogue Statistical Analysis) may be carried out in addition to corner analysis; however, we do not do so in this unit. It is sufficient, here to have an understanding of the theory without the practice.

In corner analysis, the process variable extremes are applied to a number of parameters that sum to make a particular corner. However, in Monte Carlo analysis, a statistical distribution is attributed to each parameter so that each time a model file for a particular component is accessed, the parameters will be assigned a value within their own distribution. If a number of 'runs' of the same simulation are carried out; the simulation output will be different for each. From this, a statistical performance characteristic may be deduced. In order to attribute statistical relevance to the results, the number of runs should be high (of the order of hundreds).

Two forms of statistical analysis may be applied; *process* and *mismatch*.

Process statistical analysis applies the same set of parameter values to all components of a particular model type in the design for each run (excluding, of course, the design parameters). Mismatch statistical analysis applies different values to the parameters of each device instance for each run. Process analysis may be considered to account for variation in the fabrication process between wafers (or in the extreme, dies). Mismatch analysis may be considered to account for variation in the fabrication process across the area of an individual die. For example, the Monte Carlo simulation of the performance of a (infinite output impedance) current mirror will show differing gate voltages but equal currents under process analysis, and differing gate voltage and differing currents under mismatch analysis.

10. Semiconductor Layout

This section is included as background material. Layout is not applied to the designs investigated in this unit.

Once happy with the simulated performance of the circuit, the designer is able to lay out the design on silicon by defining the areas of the different diffusions, polysilicon, oxide and metallisation in the layers of the implementation.

10.1. Component Placement

With analogue circuit layout there are some common 'good practice' measures that ensure (or, rather, promote) circuit functionality. Below is a list of the measures you would expect to find in place in commercial design.

- Keep the flow of signals and power as per the sense of the schematic cell-view except where this would lead to an uneconomical use of semiconductor area.
- Orientate all the surface layer key polysilicon features such as resistors and FET gates in the same manner.
- Only use the salicided polysilicon layer to route interconnects locally. There is no salicide in the gpdk180 design kit and so, where possible, all local routing is to be done in Metal Layer 1.
- Only use single vias and contacts when it is not possible to use multiple. For example, use two for signal lines and more for power lines where low impedance (e.g. to earth reference) or high current (e.g. power supplies) should be supported.
- Include power (V+, at the top) and ground (or V-, at the bottom) 'bus bars' to your sub-circuit layout. Make these at least wide enough to accommodate two vias to the next metal layer up.
- Lay a common N-well for pmos FETs. Lay a separate N-well for Bulk-Source connected PMOS where the Source is not connected to the positive supply.
- Make metal path widths at least 1.5 times the minimum requirement for the maximum expected current on that path.
- Shield active devices from one another. This is because of noise issues on the substrate and to increase matching reliability. Do this by encircling each with contacts to the most negative rail (for P-sub) or most positive rail (for n-well). These are known as guard rings and they also help to remove latch-up problems. It is not necessary to isolate the parts of a multiple-count FET from one another but dummy devices should be used at the ends of columns of devices to match the environment seen by each part.
- Only route each metal layer in a single orientation (either N-S or E-W) and alternate orientation between the layers. If a process does not have a salicide layer for local interconnect, metal layer 1 is used for this purpose and may route in any direction. In this unit, route even layers E-W and odd layers N-S.
- Do not route metal layers 1 or 2 over polysilicon resistors. This is because the small amount of overall capacitance between layers is increased by the generally large areas of the resistors.

- Use the higher metal layers for more global routing, i.e. within a functional block (or a layout cell-view) strive to be economical with the number of metal layers you employ, starting with those closest to the substrate.
- Add Dog Bones to the FET gates. This is an area of polysilicon abutting the end of the gates of at least twice the gate length and wide enough to accommodate a via to Metal 1. You should dimension it to accommodate at least two vias at each end of the FET so that the complete area of polysilicon resembles an 'H' shape. This is to minimise the erosion of gate area by diffraction in the photolithography process.
- Do not place vias directly onto the polysilicon over the gate area. Any gate widths greater than (say) 20 microns should be split across multiple counts and any gate widths greater than (say) 10 microns should be contacted to metal 1 inputs at both ends. This is due to the appreciable resistance of the polysilicon layer and is only an issue under high frequency operation but should always be practiced. This rule may be relaxed somewhat in processes where polysilicon includes a salicide layer.
- Do not use 'fingers' to scale transistors. This may lead to 'Length of Oxide Definition' errors. Always use 'count'.

10.2. Layout Techniques: Matching, Common Centroid and Interspacing.

The simulation processes used in the design thus far have assumed 'good' matching between components. However, this is by no means occurs naturally in the manufacture of an integrated circuit and mismatches increase as component dimensions reduce and component spatial separation increases. Additionally, matching across components of different geometries is poor and should not be expected at the design stage.

The important thing to note when laying out an analogue circuit is the symmetry and proximity of components that need to be matched. The common approaches are;

- keep matched components near to one another,
- use multiple sub-components and interspace them, and
- place multiple sub-components such that the two matched components share a common centre point in the layout.

The three methods above aim to lower the resulting performance variation caused by process variation across the die. This is known as mismatch variation. Global process variation is (mostly) inter-wafer and cannot be designed out of the circuit performance by layout and must be accounted for by **PVT**/**MC** functionality proving.

The common centre point layout technique is commonly used where non-equal component counts are required (e.g. in a 1:8 ratio current mirror). In this case multiple counts of the same sized component should always be used as the effects of dimensional tolerance should be maintained.

The reader is referred to [Hastings, *The Art of Analogue Layout*] for an in-depth address of matching techniques.

10.3. Design Checking

Two forms of check are required to ensure a correct layout. Firstly, a *Design Rule Check* (DRC) must be applied to ensure that the sizes and relative position of all the areas of meatalisation, doping, polysilicon, salicide etc. are compliant with the rules of the fabrication plant. Secondly, a *Layout Vs. Schematic Check* (LVS) must be applied to ensure that the connectivity and cell parameters are the same across the two cell-views.

In addition to the two mandatory checks above, an extraction of the parasitic reactances associated with the physical realisation of the device interconnects may be carried out and a new cell-view back-annotated with component instances. This process forms a model of the performance of the actual resulting device with the greatest accuracy. However, the number of additional components in the model is of concern to the designer as the size of the netlist will greatly increase in this operation. If left unchecked then a model of the circuit may result that is too large for simulation in a reasonable time. There is an engineering trade-off between complexity and accuracy requirements.

Commonly, extraction algorithms are limited to series resistance and parallel capacitance to ground for the device interconnects which are broken down into sections to which lumped elements are added. The parameterised models of the devices (e.g., NFET) will have sufficient accuracy already to account for the parasitic reactances associated with their realisation.

11. Process Design Kits

A Process Design Kit (PDK) contains the process technology and needed information to do device-level design in the Cadence DFII environment.

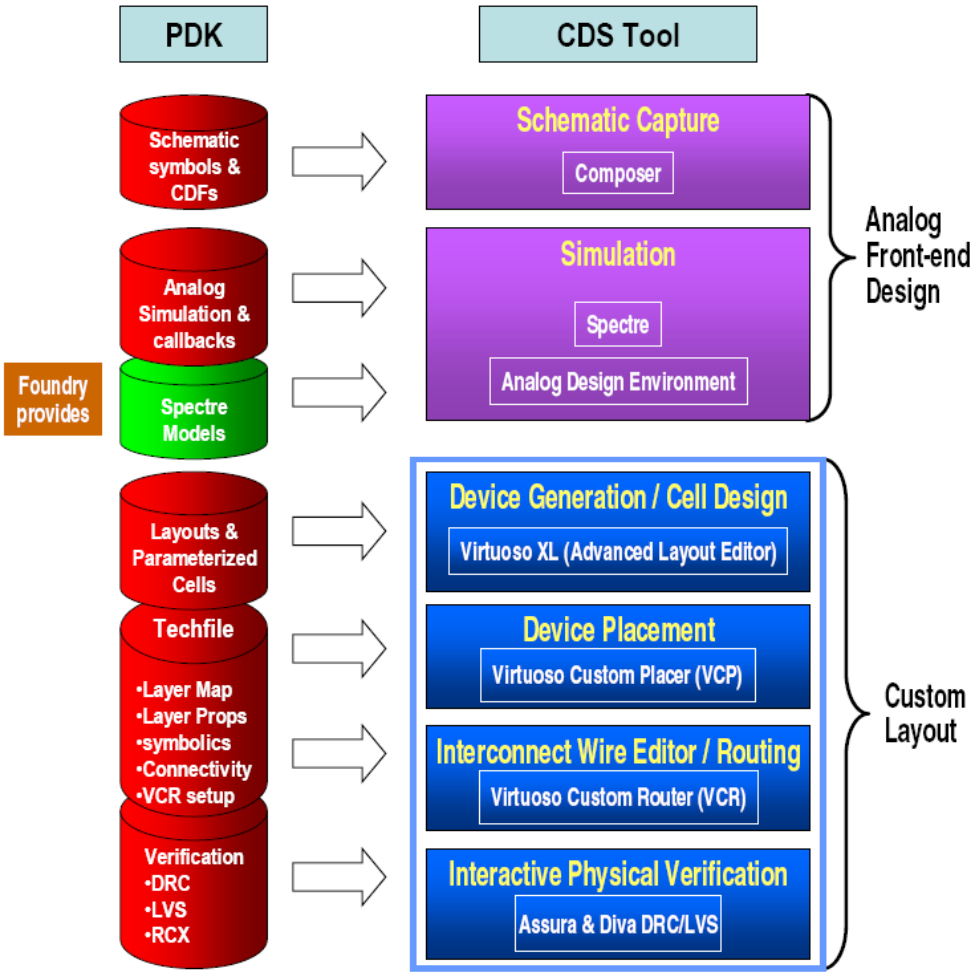


Figure 49: Contents and Use of a Process Design Kit.

The fabrication plant works with the design product vendor (In this case, Cadence) to construct the PDK. Not all fabrication plants support all design products and visa versa. Fabrication plants include

- AMI Semiconductor http://www.amis.com/
- AustriaMicrosystems http://www.austriamicrosystems.com/
- IHP http://www.ihp-microelectronics.com/
- TSMC http://www.tsmc.com/english/default.htm
- UMC http://www.umc.com/
- Jazz Semiconductor http://www.jazzsemi.com/

12. Non-assessed Laboratory Activities

Refer to the Instruction set and Videos available on a week-by-week basis for the laboratory activities.

13. Assessed Activities

13.1. Voltage Reference Circuit

13.1.1. Assessment Format and Requirements

You must use the Cadence Design Suite and the 180nm GPDK.

You should make image files of the circuit schematics and performance plots for your design and justify their content. Add to this, descriptions of the operation of your topology and justify the dimensioning of your components. The submission should be a single pdf file containing your prose, circuit diagrams and graphical outputs.

Submit your pdf file via the SAFE online system.

13.1.2. Assignment:

The task is to design an analogue system which:

- Provides a constant voltage of 1V (referred to ground) which is capable of driving a load resistance (to ground) which varies from 2kOhm to 10kOhm. This type of system is commonly called a Voltage Reference Circuit, one form of which has been discussed in the course notes. You should aim to track 1V to within 0.17% initially; additional marks are attained for increased accuracy.

To do this you are advised to use the following:

- A single-stage differential transconductance amplifier with one input from an ideal reference voltage source and the other from the system output voltage (feedback path). To increase the tracking accuracy, a two-stage op-amp may be required and the resulting high loop gain and ability to employ compensation via the Miller affect will result in additional marks.
- A NMOS transistor with its drain connected to the positive power supply and source connected to the load as an output driver stage. This should be dimensioned appropriately to meet the large output current requirement whilst keeping the differential amplifier in its high transconductance region. Such an output device requires a gate voltage range more appropriate for a single stage differential amplifier.

You may employ the following ideal components from the ‘analogLib’ library; all other components must come from the GPDK180 library.

- A DC voltage source (vdc) as the input reference; this may be of any value between ground and the power supply voltage.
- A 50uA DC current source (idc) as the bias reference for the differential amplifier (to be mirrored into your circuit)
- A 2.5V DC voltage source (vdc) as the power supply
- A ‘null’ DC voltage source (vdc) as the loop identifier for the stability analysis
- A resistor (res) as the output load
- A ground reference (gnd)
- During your design phase you may use other ideal components to test your solution but these must not remain in the final circuit.

You must produce the following analyses to show the functionality of your system:

- Single DC operating point giving annotated node voltages on the schematic circuit diagram. 40%
- DC sweep analysis with the output load resistance as the swept parameter showing the voltages at critical nodes in your system and the current into the load. 20%
- Stability analysis of the feedback system showing loop gain (in dB20) and phase. 20%
- Performance of the system over temperature (T) and process (P) variation. 20%

You are not expected to include the following

- A symbol for your circuit (i.e., all the inputs and outputs and power/ground supplies may be on the same schematic)
- Consideration of voltage (V) variation

13.1.3. Hints:

Keep the differential amplifier simple until you establish a need for additional gain and/or stability. You may not require additional stability with a single stage differential amplifier, however, the low loop gain will affect your output voltage tracking accuracy.

Remember to appropriately connect the bulk-gates of the NFET and PFET devices. This is especially important with the NFET output device.

The variable ‘temp’ is a system variable that may be controlled in a parametric analysis; temperature may also be selected in the DC sweep form.

There are marks available for the correct application of the design flow as well as achieving a working solution.

14. 180nm Generic PDK Used in This Unit

The design kit used in this unit is generic. It is non-manufacturable yet represents the state-of-the-art in process technology. *180nm* refers to the process node. It is not necessarily the minimum FET gate length as is commonly assumed.

The remainder of this section summarises the design kit contents. The full reference manual is available in the model library within the cadence software.

14.1. Supported Devices

14.1.1. MOSFETS

nmos - NMOS transistor

nmos3 - 3 terminal NMOS transistor with an inherited connection for the body

pmos - PMOS transistor

pmos3 - 3 terminal PMOS transistor with an inherited connection for the body

In addition to the devices above, high frequency versions (nmosrf, pmosrf) are available.

The most commonly used MOFET devices in this unit are the high voltage versions of the above; nmoshv and pmpshv. These devices support voltages up to 2.5 Volts.

14.1.2. Resistors

nplusres – N+ Diffused Resistor	Rho = 300 ohms/sq
polyres – Poly Resistor	Rho = 7.5 ohms/sq

14.1.3. Capacitors

mimcap - metal-to-metal capacitor	CapA = 1.0fF/um^2
nmoscap - NMOS transistor configured as a capacitor	CapA = 8.0fF/um^2

14.1.4. Inductor

inductor – Metal 3 Inductor

14.1.5. BJTs

vpnp - CMOS vertical PNP with substrate collector: Fixed emitter area

npn - Bipolar NPN: Variable emitter area for layout, fixed emitter area for simulation

pnp - Bipolar vertical PNP: Variable emitter area, fixed emitter area for simulation

14.1.6. Diodes

ndio - N type diode: Variable diode length and width

pdio - P type diode: Variable diode length and width

14.2. CDF (Component Description Format) Parameters

The CDF parameters are used to define the physical realisation of the devices. The device models use these parameters as inputs in the circuit simulation.

14.2.1. MOSFETS

Model Name	Spectre model name (non-editable)
l (M)	Gate length in meters
Calculate Width Method	Cyclic which controls user entry of total Width value or individual finger Width value
w (M)	Gate width in meters (non-editable using fingerWidth)
Number of Fingers	Number of poly gate stripes used in layout (w/nf width)
Width Per Finger	Width of each gate stripe (non-editable using totalWidth)
Multiplier	Number of Parallel MOS devices
Calc Diff Params	Cyclic which controls the calculation of area and periphery of the source/drain regions for simulation. Default is true which auto calculates the values. If nil, the user can enter the values.
Source diffusion area	Calculated source diffusion area in square meters
Drain diffusion area	Calculated drain diffusion area in square meters
Source diffusion periphery	Calculated source diffusion periphery in meters
Drain diffusion periphery	Calculated drain diffusion periphery in Meters

14.2.2. Resistors

Resistance (ohms)	Resistance value used for simulation
Contact Resistance (ohms)	Resistor contact resistance in ohms
Width (M)	Resistor width in meters
Length (M)	Resistor length in meters
Entry Mode	Allows user to specify either “r and w” or “l and w”
Multiplier	Number of Parallel Resistor devices

14.2.3. Capacitors

Capacitance	Capacitance value used in simulation
total Capacitance	Capacitance multiplied by Multiplier (non-editable)
l (M)	Capacitor length in meters (non-editable for mimcap)
w (M)	Capacitor width in meters (non-editable for mimcap)

MIMCAP: The following parameters are for the mimcap only.

Spec	Cyclic used to choose capacitor entry method (Capacitance, Cap & w, l & w)
CapA (F/M^2)	Plate Capacitance (non-editable, units are Farads per Meter Squared)
CapP (F/M)	Fringe Capacitance (non-editable, units are Farads per Meter)

NMOSCAP: The following parameters are for the nmoscap only

Spec	cyclic used to choose capacitor entry method (Capacitance, Cap & l, Cap & w, l & w)
Width Per Finger	Width of each gate stripe (non-editable using totalWidth)
Number of Fingers	Number of poly gate stripes used in layout (w/nf width)
Multiplier	Number of Parallel MOS devices

14.2.4. Inductor

Inductance (H)	Inductance in Henry's based on calculation (non-editable)
Number of Turns	Fixed number of coil turns (1.5, 2.5, 3.5, 4.5, 5.5, 6.5, 7.5, 8.5, or 9.5)
Inner Radius	Radius of inner coil (non-editable)
Inductor Space	Space of top metal (non-editable)
Inductor Width	Width of top metal (non-editable)

14.2.5. BJTs

Model name	Spectre model name (non-editable)
Area	Emitter Area (non-editable)
Emitter width	The width of the emitter, in meters (For npn and pnp only)

Emitter Size	The length and width of the emitter, in meters (For vpnp only)
Multiplier	Number of Parallel Bipolar devices

14.2.6. Diodes

Model name	Spectre model name (non-editable)
Device Area	Calculated junction area in meters squared (non-editable)
Multiplier	Number of Parallel Diode devices
Length (M)	Diode length in meters
Width (M)	Diode width in meters

14.3. Device Parameter Ranges

14.3.1. MOS Parameter Ranges (we use hv versions in this unit)

DEVICE	l (gate length in microns)	
	min	max
nmos	0.18	20
pmos	0.18	20
nmos3	0.18	20
pmos3	0.18	20
DEVICE	w (gate width in microns)	
	min	max
nmos	0.42	100
pmos	0.42	100
nmos3	0.42	100
pmos3	0.42	100
DEVICE	Nf (Number of Fingers)	
	min	Max
nmos	1	100
pmos	1	100
nmos3	1	100
pmos3	1	100

14.3.2. Resistor Parameter Ranges

DEVICE	r (resistance in ohms)	
	min	max

nplusres	1.5
polyres	0.135

DEVICE	w (resistor width in microns)	
	min	max
nplusres	0.6	100
polyres	0.6	50

14.3.3. Capacitor Parameter Ranges

DEVICE	c (capacitance in farads)	
	min	max
mimcap	17.6f	912f
nmoscap	0.12f	444f

DEVICE	w (capacitor width in microns)	
	min	max
mimcap	4	30
nmoscap	0.6	100

DEVICE	l (capacitor length in microns)	
	min	Max
mimcap	4	30
nmoscap	0.6	20

DEVICE	fingers (number of capacitor fingers)	
	min	Max
nmoscap	1	50

14.3.4. Inductor Parameter Ranges

DEVICE: Inductor	min	max
ind (inductance in Henrys)	2.09E-10	4.43E-8
Nr (inductor turns)	1.5	9.5
Rad (inductor inner radius in microns)	20	100
Space (inductor metal space in microns	2	10
Width (inductor metal width in microns)	5	20

14.3.5. BJT Parameter Ranges

DEVICE	emitter size (fixed emitter size in microns)	
	min	max
vpnp	1.3*13	1.3*13

DEVICE	w (variable emitter width in microns)	
	min	max
npn	0.6	1
pnp	0.6	1

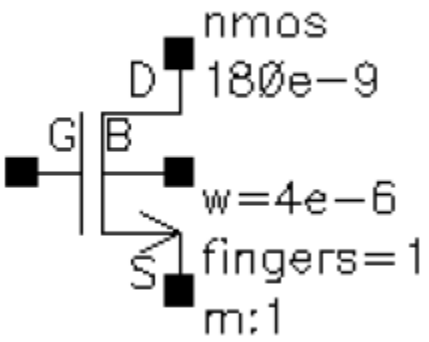
14.3.6. Diode Parameter Ranges

DEVICE	l (diode length in microns)	
	min	max
ndio	0.6	20
pdio	0.6	20
DEVICE	w (diode width in microns)	
	min	max
ndio	0.6	20
pdio	0.6	20

14.4. Device Schematics

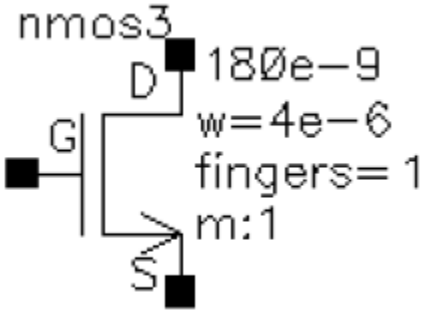
14.4.1. nmos – NMOS Transistor

We generally use the high voltage (nmoshv) version in this unit.



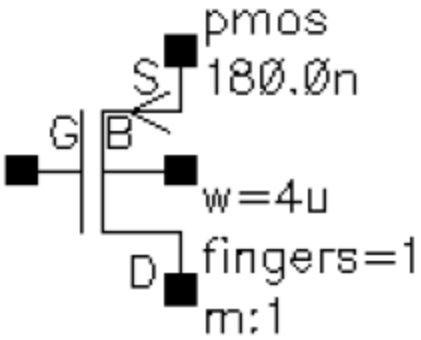
14.4.2. nmos3 – NMOS Transistor

We avoid the use of this component in this unit.



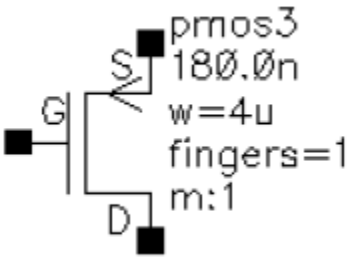
14.4.3. pmos – PMOS Transistor

We generally use the high voltage (pmoshv) version in this unit.

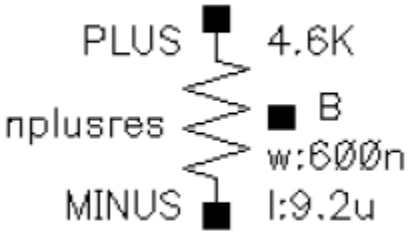


14.4.4. pmos3 – PMOS Transistor

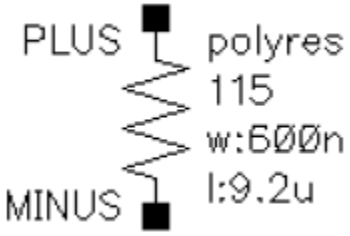
We avoid the use of this component in this unit.



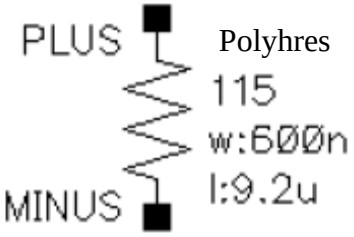
14.4.5. nplusres – Nplus Resistor



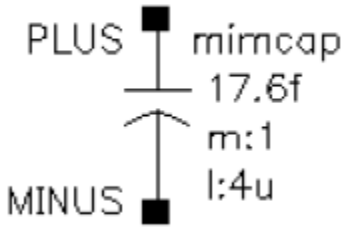
14.4.6. polyres – Polysilicon Resistor



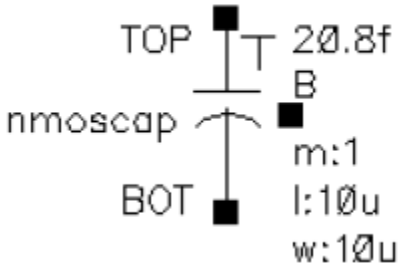
14.4.7. polyhres – High Resistivity Polysilicon Resistor



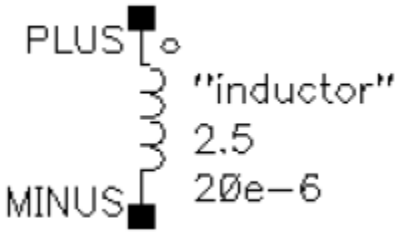
14.4.8. mimcap – Metal-to-metal Capacitor



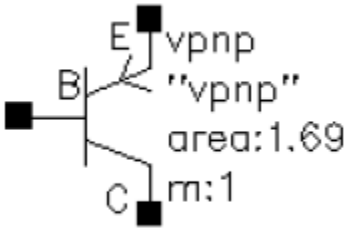
14.4.9. nmoscap – Nmos Transistor Configured as a Capacitor



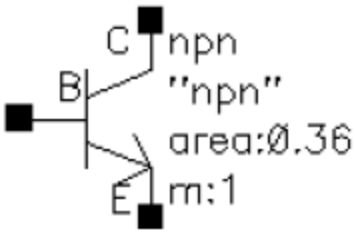
14.4.10. inductor – Metal 3 Inductor



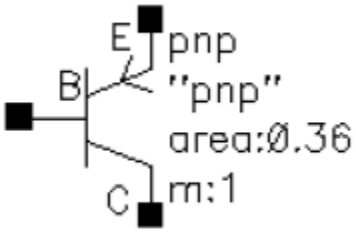
14.4.11. vpnp – Vertical Bipolar PNP Transistor



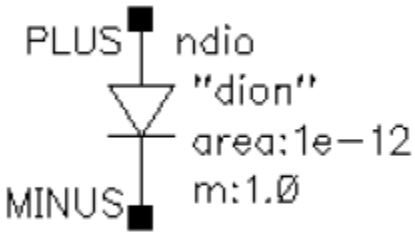
14.4.12. npn – Biploar NPN Transistor



14.4.13. pnp – Bipolar PNP Transistor



14.4.14. ndio – N type Diode



14.4.15. pdio – P type Diode

